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Department of Bioengineering

Nail-Mounted Haptic Cues for Piano Training and Tele-training

Author:
Sheng Jiang

Supervisor:
Dr. Alexis W.M. Devillard

Co-Supervisor:
Dr. Etienne Burdet

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Abstract

Skilled hand movement depends on choosing the correct finger as well as the correct target, yet guidance technology poorly supports that choice. Screens and sounds show which key to press and when, but neither directly specifies the acting digit. MIDI likewise records the key and timing, not the finger used, so compliance is not routinely measurable. This work addresses both gaps. A platform delivered vibration to the nail of the target finger while leaving the fingertip free to sense the key, with per-key illumination and camera-based finger identification aligned to MIDI. Bench tests selected the actuator, mounting, and drive settings before recruitment. A constraint-driven generator then produced sequences of measured, ordered difficulty matched across conditions. Twenty participants performed the task with key-only, visual-finger, and vibrotactile-finger guidance. Relative to matched visual cues, vibrotactile cues reduced reaction time by 247 ms and improved finger accuracy by 3.65 percentage points. Benefits varied across digits, and wrong-hand errors fell from 28.8% to 2.7% of substitutions, while key accuracy remained near ceiling in both conditions. The digit and error patterns suggest that the cue primarily aided selection of which finger should act rather than its movement. The common key–finger representation also drove a working remote-teaching deployment supporting live and recorded guidance. The evidence establishes an immediate cue-guided performance advantage within one session; it does not establish retention after cue withdrawal or behavioural benefit during remote use.

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Chapter 1

Introduction

1.1 Motivation and Related Work

Keyboard performance is a demanding sensorimotor task: within a fraction of a second, the actor must identify a target key, assign a finger, and execute the movement while the cue–action mapping is still forming [1]. Fingering determines whether the hand reaches a posture from which the next note is accessible, so a poor choice can propagate through a sequence. It is also difficult because the digits are neither mechanically nor neurally independent; passive coupling particularly limits the index, middle, and ring fingers [2, 3]. Selecting a specified digit recurs in fine-motor training and rehabilitation, where keyboard practice and repetitive finger tasks provide structured, quantifiable exercise [4, 1, 5]. This project therefore uses the keyboard as an instrumented model of finger-specific guidance rather than musical achievement: it discretises responses, timestamps events, and makes the intended finger scorable.

Augmented feedback can improve motor performance, but its channel determines what it can express, and an immediate benefit does not necessarily imply learning [6, 7]. Visual displays suit spatial targets, and auditory feedback conveys timing or outcome; neither directly specifies the acting digit, so the user must infer a fingering or decode a symbolic label within a sensorimotor loop governed by predicted and actual sensory consequences [8, 9]. A bodily haptic cue can instead specify the effector directly. Vibrotactile stimulation of novice pianists has produced lower timing error than visual stimulation, although note-accuracy evidence was unresolved [10]; comparisons with electrical stimulation and linkage-based devices reveal modality-specific trade-offs rather than a universal winner [11]. Explicit error feedback can also overcome performance ceilings that repetition alone does not [12]. These findings motivate haptics, but do not establish whether a finger-addressed cue outperforms matched visual finger information.

Screen tutors, mobile applications, games, and virtual environments have made keyboard practice accessible and measurable [5, 13, 14, 15], yet typically show what to press without specifying which finger should press it. The omission becomes sharper remotely: a co-located teacher can reposition the learner’s hand, whereas networked instruction must encode finger identity through the available channels. This work supplies that missing channel locally and carries it into a remote implementation.

Two limitations remain. The first is cue placement. Most wearable haptics in this domain signal timing, error, or a global prompt at hand or body level rather than digit identity [11]. Per-finger actuation requires a site that conveys the cue without disrupting the trained action. Although the fingertip has the hand’s highest tactile innervation density [16], an actuator there obstructs the surface sensing key edge, contact, and release. Wearable designs therefore move actuation off the pad, including proximal-phalanx, dorsal, and passive-learning systems [17, 18, 19, 20, 21].

The fingernail offers a rigid, repeatable mounting surface close to the pad but outside its contact patch. It supports measurable tactile sensibility and localisation across multiple stimulation sites [22, 23]; nail-worn and fingerpad-free devices also establish the feasibility of this form factor [24, 25, 26]. Stimulation might additionally be referred towards the fingertip, placing the percept near the action while leaving contact physically free, but referral during active key contact is not established here or in the cited literature. It remains a design rationale, not a claimed platform property.

The second limitation is measurement. MIDI records which key was pressed and when, not which finger pressed it. Finger-guidance studies must therefore infer compliance from the key sequence or use motion capture and instrumented gloves that may load the hand. The literature makes finger-specific haptic guidance plausible, but leaves it neither implemented at ten-finger resolution on a non-occluding site nor evaluated with finger-resolved action data. A cue that cannot be scored against the requested action cannot be adequately tested.

1.2 Aim, Research Questions, and Objectives

This project addresses that combined design and measurement gap. It aims to build and evaluate a multimodal keyboard-guidance platform that delivers finger-specific vibration from nail-mounted actuators while preserving natural key contact; compare it with matched visual finger cues under controlled bimanual sequence difficulty; and carry the same key–finger representation into tele-training.

- **RQ1.** When target-finger information is matched, do nail-mounted vibrotactile and visual finger cues differ in reaction time and finger-specific action accuracy under matched sequence difficulty?
- **RQ2.** Does any difference vary across homologous fingers 1–5, and do the modalities produce the same kinds of finger error?

The objectives are to **(O1)** integrate MIDI, illuminated keys, nail actuators, embedded control, camera finger recognition, and synchronised logs; **(O2)** validate actuator, attachment, and drive choices; **(O3)** generate seeded bimanual sequences with measurable, ordered difficulty matched across conditions; **(O4)** compare key-only, visual-finger, and vibrotactile-finger guidance within participants, overall and by digit; and **(O5)** implement synchronous and asynchronous tele-training in which a teacher’s recognised key–finger events are reissued through the learner’s illumination, screen, and actuators.

The keyboard is treated as a spatial response surface and sequences are generated, so prior repertoire and auditory memory cannot substitute for the cue [27, 28]; musical expression is outside scope. The work studies finger-choice cueing, not rendering of key mechanics. Its single-session evidence concerns immediate guided performance, not retention after cue withdrawal. Tele-training is a platform deployment rather than an experimental condition: network guidance was not manipulated and received no user study, so no behavioural claim is made about remote delivery.

Chapter 2

Methods

2.1 System Implementation

The platform (Figure 2.1) uses a Nektar SE25 MIDI keyboard with 25 calibrated keys (notes 48–72). Two under-key WS2812 strips cue the target key without altering key feel; ten nail-mounted actuators cue the fingers. A Teensy 4.1 and two PWM boards drive the actuators from a motor supply isolated from logic power.

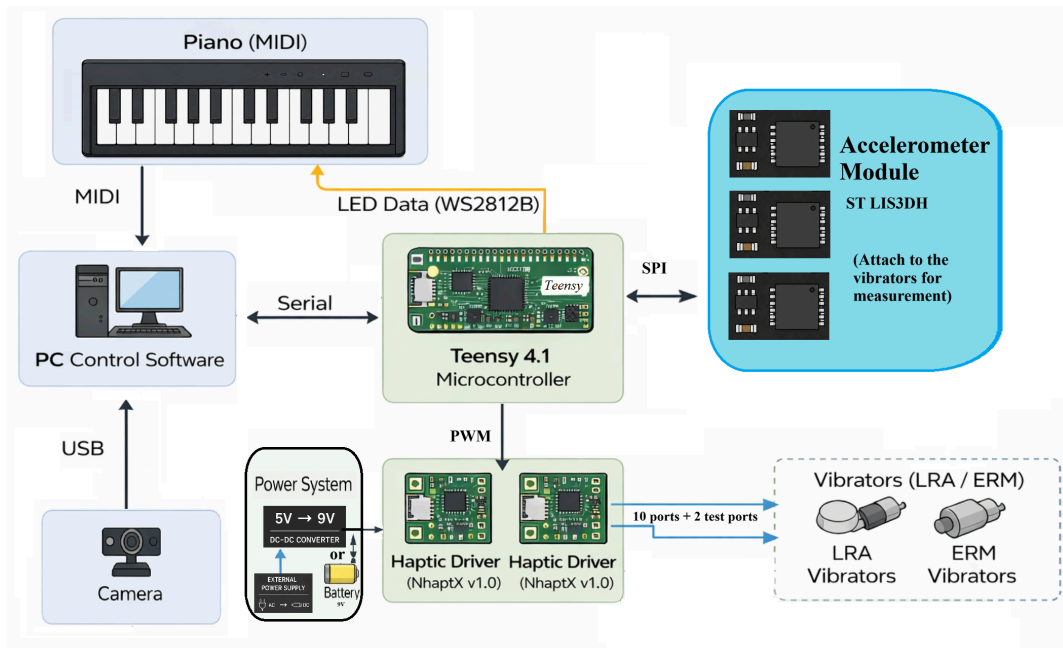


Figure 2.1: Hardware architecture. The host dispatches LED and nail-actuator cues through the Teensy while MIDI and video return on separate links. Twelve motor channels provide ten finger outputs and two test/spare outputs. The 9 V actuator rail is supplied by a battery or, as here, a converter from an external 5 V supply. Accelerometers were coupled only for the bench tests in Section 2.2 and were absent from participant sessions.

Python software schedules trials, drives cues, logs MIDI, and records 1280×720 , 30-fps video. All streams receive Unix timestamps from the host clock. Appendix Table C.1 records the exact hardware and software configuration. The integrated electronics and exposed bench and session interfaces are shown in the 3D-printed enclosure in Appendix Figure C.1.

2.2 Hardware Validation and Final Configuration

Three pre-recruitment bench experiments fixed the configuration. A coupled accelerometer measured demeaned three-axis vector RMS. Frequency was set before amplitude, giving a fixed 224-Hz, 25.1%-duty PWM operating point. The selected duty was the measured point closest to a provisional target band rescaled from an earlier project metric; it defines a reproducible setting, not a psychophysically calibrated intensity [29]. LRA and ERM onset latencies were compared over ten trials each using CUSUM detection [30]. Double-sided tape, Blu Tack, and eyelash adhesive were compared over three independent five-trial mounts, with mount as the averaging unit. Only transmitted intensity was measured; handling was judged rather than scored. The rig, sweeps, selection calculation, detection settings, and attachment metrics are in Appendix B.

2.3 Main User Study

2.3.1 Integrated Operation and Synchronisation

Each event illuminates the target key and, where applicable, displays or vibrates the target finger. The visual finger cue was shown on a small display standing immediately above the keyboard, in the position sheet music would normally occupy. It sat close enough that the cue, the keys, and the participant’s hands remained within a single field of view, so the cued finger could be read without looking away from the keys. The dispatch callback timestamps cue onset. The first MIDI note-on ends the event; an incorrect press is therefore an error, not a retry. Events time out after 5 s and are separated by 400 ms.

Camera start-up latency prevents direct conversion from host time to frame number: the first-frame offset varied by roughly -0.2 to $+0.55$ s, or up to sixteen frames. A keyboard LED flashed after recording starts supplies both a host timestamp and video frame, re-anchoring every trial. Pulse-duration and edge-agreement checks reject a false anchor; the complete mapping is in Appendix B.

2.3.2 Controlled Stimulus Sequence Generator

To avoid repertoire familiarity and auditory or rhythmic confounds [27, 28], stimuli followed a constrained serial-response design [31, 32]. Each sequence contains 30 single actions (one hand, finger, and key), with both hands represented. Keys are drawn from 15 calibrated white notes; black keys were excluded because their row changes reach and posture [33]. Difficulty is the unweighted vector $\mathbf{D} = (C_m, C_s, C_c)$: motor cost, sequence complexity, and bimanual coordination. Candidates must satisfy every level-defining range in Table 2.1.

Table 2.1: Level-defining constraints; ranges using the note span S are normalised by it. All levels use both hands. Overlap ratio r is 0, 0.15, 0.30 for α, β, γ ; only γ permits crossing.

Level	Motor and sequence constraints	Coordination constraints
α	$\bar{d}^{\text{seq}}/S \leq 0.12, \bar{d}^m/S \leq 0.10, \bar{d}_{95}^m/S \leq 0.16, \bar{d}^f \leq 1.0, R_h/S \leq 0.30, 0.15 \leq H_{\text{norm}} \leq 0.35.$	$0.15 \leq A_h \leq 0.35, B_h \geq 0.65, O_{LR} = 0, X_f = 0, X_e = 0.$
β	$0.08 < \bar{d}^{\text{seq}}/S \leq 0.22, 0.08 < \bar{d}^m/S \leq 0.20, \bar{d}_{95}^m/S \leq 0.30, \bar{d}^f \leq 2.0, R_h/S \leq 0.45, 0.40 \leq H_{\text{norm}} \leq 0.65.$	$0.30 \leq A_h \leq 0.55, B_h \geq 0.70, 0 < O_{LR} \leq 0.15, X_f = 0.$
γ	$0.15 < \bar{d}^{\text{seq}}/S \leq 0.38, 0.15 < \bar{d}^m/S \leq 0.35, \bar{d}_{95}^m/S \leq 0.45, \bar{d}^f \leq 3.0, R_h/S \leq 0.65, 0.65 \leq H_{\text{norm}} \leq 0.90.$	$0.50 \leq A_h \leq 0.80, B_h \geq 0.75, 0.10 \leq O_{LR} \leq 0.30, 0 < X_f \leq 0.20, X_e/S \leq 0.20.$

Seeded rejection sampling produces reproducible pools, then family matching makes sequences within each level comparable. Before locking, level medians had to follow the intended ordering with fewer than 10% adjacent-level violations. Transition uncertainty and displacement also have independent grounding in choice-reaction accounts [34] and Fitts’ law [33]. Definitions, rejection rules, and matching tolerances are in Appendix B.

2.3.3 Camera-Based Finger Identification

Because MIDI identifies the key and time but not the acting finger, calibrated key regions and the two-hand MediaPipe landmark model [35] supply finger evidence without markers, gloves, or depth sensors. An edge-constrained calibration follows each visible key shape instead of assuming rectangles, and assignment uses the key actually pressed rather than the cued target. For visible fingertips $\mathcal{V}_t$, pixel distances $d_{f,t}$ to that key become a softmax distribution [36, 37]:

$$p_t(f) = \frac{\exp(-d_{f,t}/\tau)}{\sum_{f' \in \mathcal{V}_t} \exp(-d_{f',t}/\tau)}, \quad F_t = I(p_t(f_t) \geq \theta).$$

The target finger is correct when its mass reaches θ . This graded rule avoids converting a near-tie into participant error merely because another fingertip is marginally nearer. Figure 2.2 shows the evidence on one review frame. Before data collection, $\tau = 25$ px and $\theta = 0.40$ were fixed against key width and near-tie behaviour (Appendix B).

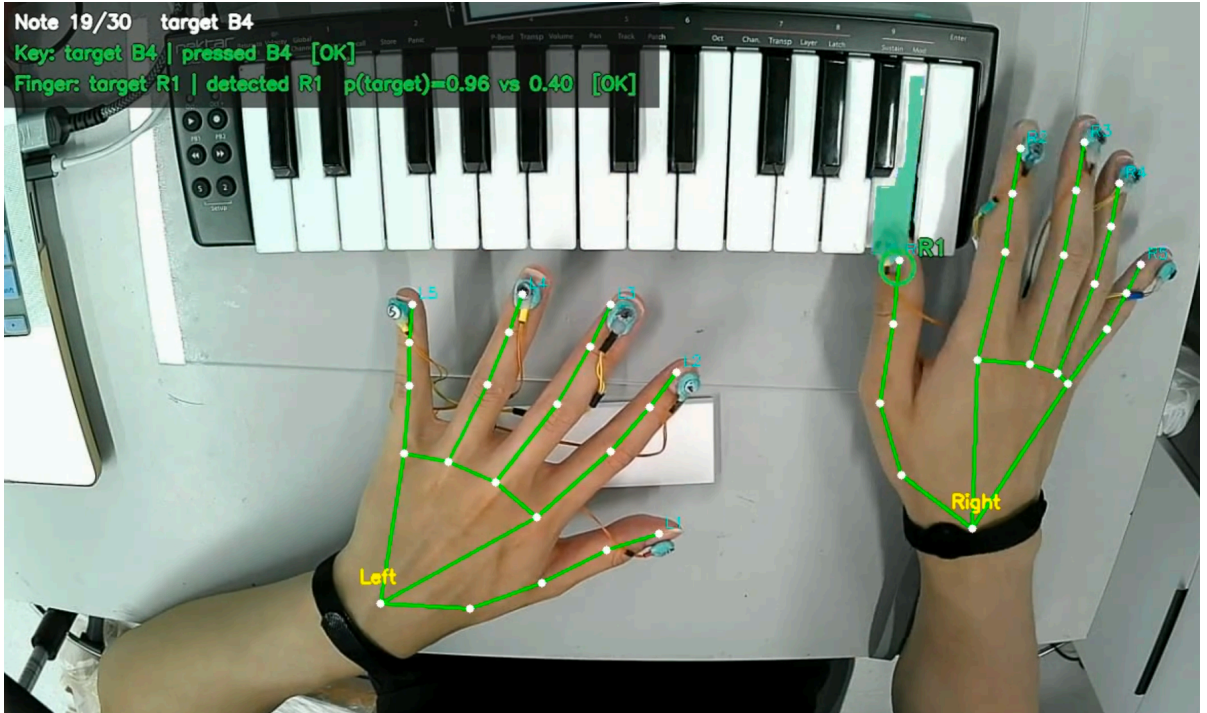


Figure 2.2: Finger identification on a review frame: MediaPipe landmarks with labelled fingertips, the pressed key’s region shaded, and the overlay reporting the target finger’s mass against $\theta = 0.40$.

Automated assignment was only a pre-screen. One reviewer inspected every event frame by frame, overrode incorrect labels, and left occluded events unresolved. Thus every scored event has visual evidence, although no second rater cross-checked the labels.

2.3.4 Design and Procedure

Twenty participants completed the study (19 male, one female; mean age 22.9 years, SD 2.2, range 20–28; 16 right- and four left-handed). Sixteen reported no keyboard experience and four reported 1–10 years; experience and handedness were metadata, not eligibility criteria or analysis factors.

The fully within-participant design crossed three feedback conditions (Figure 2.3) with three difficulty levels and three distinct sequences: 27 trials and 810 target events per participant. Every condition backlit the target key. Participants wore the actuators throughout, so only finger-cue availability and modality changed. LED output was fixed at no less than 3:1 contrast [38]. Condition C vibrated continuously until keypress or timeout, matching the visual cue’s envelope rather than adding pulse rate as a perceptual variable [39]. No auditory feedback was presented during the main user study: keyboard sound and all other audio output were muted to remove auditory-motor feedback and acoustic interference [27, 28].

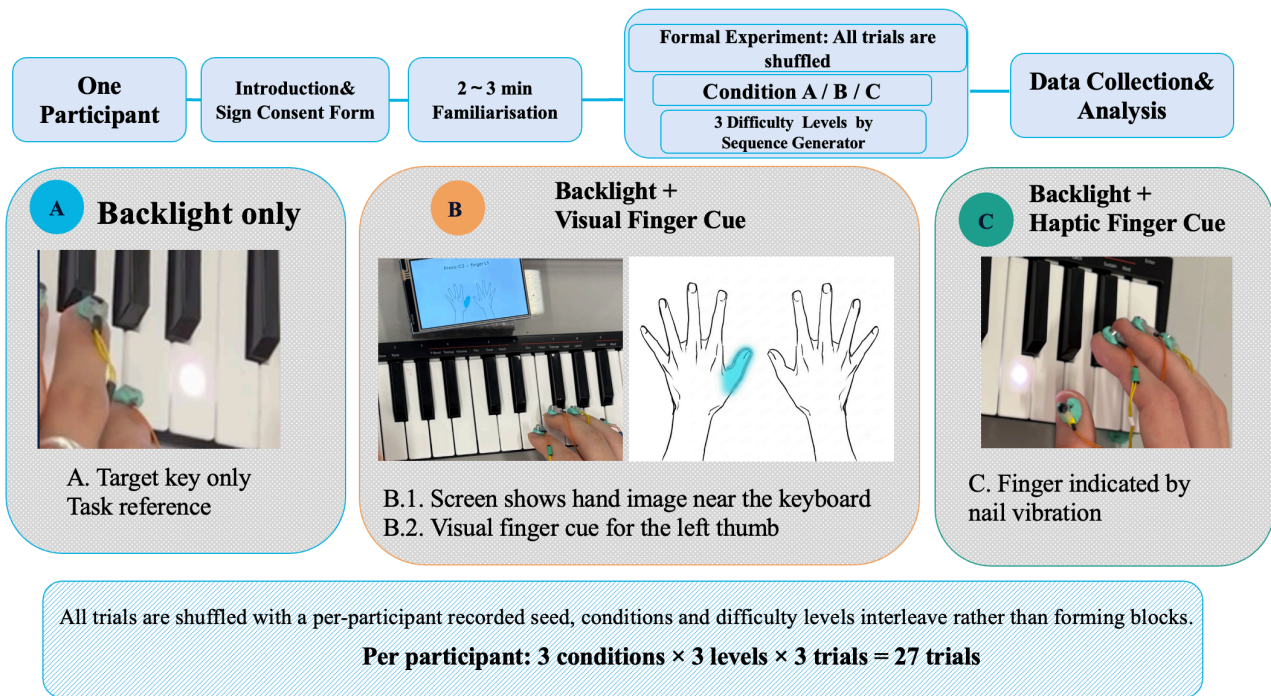


Figure 2.3: Main user study: session flow (top), the three feedback conditions (middle), and the $3 \times 3 \times 3$ trial structure (bottom). Actuator attachment and the ten-finger identification check precede familiarisation and are not drawn as separate steps.

Conditions B and C both specify the target finger and differ only in visual versus nail-vibration delivery. Condition A gives no finger cue; its agreement with the hidden target is an unguided-choice reference, not a modality baseline.

The trial list was pre-built and shuffled with a recorded participant-specific seed, interleaving condition and difficulty rather than blocking them. A randomised ten-finger check had to be completed without error before testing. Bench height was adjusted per participant [40, 41]. Between trials, participants could request a self-paced break of up to 5 min.

2.3.5 Outcome Measures and Analysis

For event t , MIDI gives key correctness $K_t = I(\hat{m}_t = m_t)$, the threshold gives finger correctness F_t , and $A_t = I(K_t \wedge F_t)$ denotes a complete action. Primary outcomes are $FA_{\text{main}} = \sum_t A_t / 30$ and cue-to-keypress reaction time for $A_t = 1$ events; every ANOVA uses this reaction time. Auxiliary $FA_{\text{cond}} = \sum_t A_t / \sum_t K_t$ isolates finger selection among correct-key responses, while correct-key reaction time, error composition, timeouts, and no-majority fingertip events check robustness and data quality.

Measures were aggregated by participant, condition, level, sequence, and repetition; the participant remained the unit of inference and event counts were never treated as independent observations. Paired B–C differences in reaction time and FA_{main} use 95% participant-level bootstrap confidence intervals, Cohen’s d_z , and Wilcoxon signed-rank checks; the same percentile-bootstrap interval is used for graphical uncertainty around group means, so a near-ceiling accuracy interval cannot cross 0 or 100%. A repeated-measures reaction-time ANOVA crossed Condition (B, C) with Finger ID (1–5), using Greenhouse–Geisser correction for Finger and the interaction because inter-digit coupling differs [2]. Secondary models replaced Finger with difficulty and repetition; both were subordinate to the pre-specified finger model and were not multiplicity-corrected. Accuracy was excluded from ANOVA because it is bounded and ceiling-compressed; participant-level differences, per-finger cells, and sensitivity across $\theta \in \{0.30, \dots, 0.50\}$ are reported instead. The ceiling rules out the F ratio rather than every model, so accuracy was additionally fitted as a binomial mixed model: cell successes over judged events under a logit link, Condition $\times$ Finger ID as fixed effects and a participant random intercept, estimated by maximum likelihood with adaptive Gauss–Hermite quadrature, tested by likelihood ratio and reported as odds ratios with participant-clustered robust intervals on a $t(N - 1)$ reference. The random intercept is what keeps events from entering as independent observations, and the model is a sensitivity analysis: the participant-level paired contrast remains the pre-specified inference, and the fit is reported in order to show whether that conclusion depends on the aggregation. Where inferential results are displayed on figures, $*p < 0.05$ and $**p < 0.01$; unmarked effects do not meet the 0.05 threshold or the panel is explicitly descriptive. Each star uses the same test reported in the corresponding text and caption.

An exploratory ten-finger model combined habitual, motor, and cue evidence using leave-one-participant-out folds and a multinomial-logistic benchmark (Appendix C.1).

2.4 Tele-training Deployment

The key–finger representation was extended to teacher and learner clients connected through a relay server (Figure 2.4). Each station was physically self-contained: its keyboard, camera, and cue hardware had no direct connection to the other station, and the two computers exchanged events only through the network. In synchronous mode, the teacher’s played key and camera-identified finger are reproduced by the learner’s key light, screen, and actuators. Only the learner wears the nail-mounted actuators because the teacher generates rather than receives the finger cues. In asynchronous mode, a recorded plan is downloaded and scheduled locally.

Figure 3.1 shows the complete paired configuration, in which a divider and headphones limited visual and acoustic cross-talk. The main study used only the student station, with audio muted.

Reaction time starts at the learner’s local cue-ready instant, so network delay shifts cue presentation but cannot enter the interval. Visual and haptic components are dispatched together locally.

This engineering demonstration, not a participant experiment, used 1,000 transport probes, alignment data from eleven two-machine sessions, and two single-operator modality demonstrations. Protocol and clock conventions are in Appendix B.

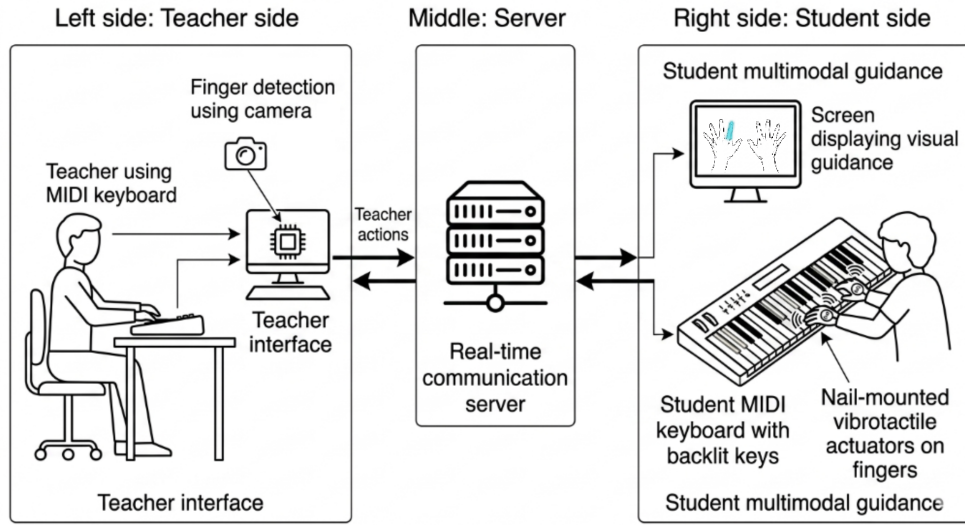


Figure 2.4: Tele-training architecture. Key–finger events cross the relay; cue timing and multimodal output are generated on the learner machine.

2.5 Ethics, Consent, and Data Handling

2.5.1 Approval, Consent, and Participation

The work was conducted under Imperial College approval for “Cross-modal platform to investigate multisensory integration during human machine interaction” (approval number: 22IC7699), granted before recruitment. Participants received the purpose and procedure in advance and gave written consent for participation and data use. Participation was voluntary, with withdrawal permitted without penalty; there was no dependent relationship, coercive incentive, or deception.

2.5.2 Data Protection

Data are pseudonymised rather than anonymous: datasets and videos use participant codes, while the identity link is stored separately on secure institutional storage, consistent with the Data Protection Act 2018 [42]. Recordings exclude faces. Privacy and validity conflict because frame-level verification cannot be reconstructed from landmark streams alone; raw hand video is therefore retained only for the approved period under restricted access. Tele-training sends cue events over TLS-protected HTTPS and secure WebSocket, and the relay stores no video [43, 44].

2.5.3 Participant Safety

Standards for hand-transmitted vibration assume sustained power-tool contact at the grip and do not establish safety for brief, high-frequency transmission through the nail [45, 46]; quoting their exposure margin would give false assurance. The fixed calibrated amplitude, event-limited cues, bounded protocol, and monitoring for discomfort, warmth, irritation, or detachment controlled exposure; sustained stimulation and adaptation were avoided [47]. No adverse event occurred, but this observation does not establish safety for longer use, which requires a dose study. Adhesive contacted only the nail, putty was discarded between participants, and detachment

was a stopping condition for both safety and cue validity.

2.5.4 Scientific Integrity and Institutional Responsibility

Automated finger labels were treated as a pre-screen: every event was reviewed frame by frame and corrected where necessary. Seeded, locked stimuli, pre-fixed parameters, threshold sensitivity analysis, and reporting of ceiling compression, unmatched physical onset, and the small key-accuracy difference alongside favourable results protect against selective analysis. Code, materials, and parameter provenance support reproduction, challenge, and handover (Appendix B). No animal work was involved.

2.5.5 Stakeholders, Justice, and Wider Impact

Stakeholders include learners, teachers, rehabilitation users [5, 48], researchers, and the institution. The small sample (20 people aged 20–28; 19 male) cannot support generalisation across age, sex, nail geometry, or hand morphology. Requiring nail-vibration perception and two-hand keyboard positioning may exclude people with neuropathy or impaired hand function, including intended rehabilitation users. The visual cue met 3:1 contrast [38]; components are commercially available, but affordability was not measured. Reusable electronics limit single-use waste, while disposable adhesive protects hygiene; no lifecycle or environmental assessment was performed.

Continuous hand video can become surveillance, so deployment should favour local processing or a non-video mode where possible. Future users should retain control over recordings and derived performance data rather than transferring ownership implicitly to a platform provider. Claims must also remain limited to immediate, single-session guided performance, not learning, retention, or clinical benefit. These choices enact the Engineering Council principles [49]: rigorous and auditable measurement, honesty about evidential limits, respect for law and public welfare, and reproducible documentation that permits criticism and extension.

Chapter 3

Results

3.1 Platform and Hardware Validation

The paired tele-training configuration exchanged key–finger events through the network relay; the teacher generated cues and wore no actuators, while the divider and headphones limited cross-talk. The main study used only the student station on the right (Figure 3.1), completing 540 trials across 20 participants without protocol deviation while logging MIDI and video for offline scoring.

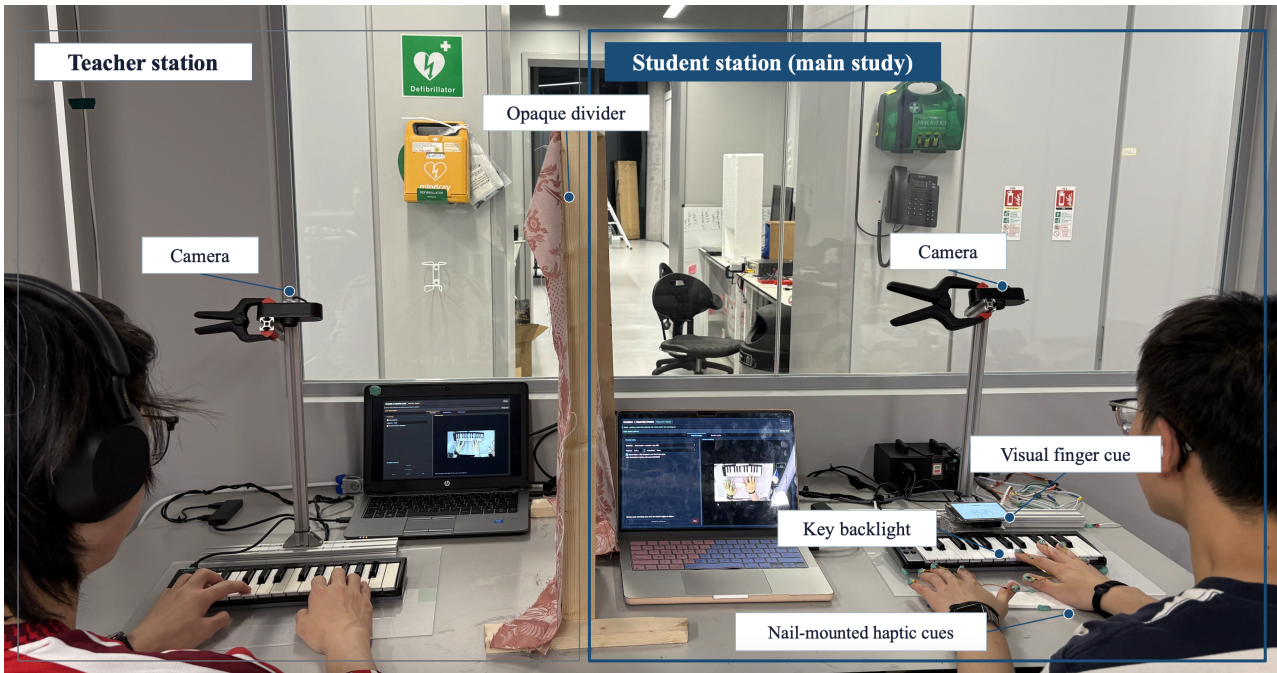


Figure 3.1: Annotated platform layout. The grey outline marks the teacher station (left), and the blue outline marks the student station used for the main study (right). Both stations use an overhead camera; the student station provides the screen-based visual finger cue, key backlighting, and nail-mounted haptic cues. The central opaque divider blocks the operators’ direct line of sight. The two-person arrangement shows the complete tele-training platform; **the main user study used only the blue-outlined student station.**

3.1.1 Actuator Characterisation and Drive Calibration

The ERM output rose broadly with duty, whereas the LRA concentrated its output in a narrow frequency band, allowing intensity to vary at fixed frequency. A frequency sweep located the mounted resonance at 224 Hz. At

that frequency, the PWM-amplitude sweep selected the firmware parameter $\text{amp}_{\text{PWM}} = 64$ on its 0–255 scale. This parameter sets the duty cycle, $D = \text{amp}_{\text{PWM}}/255 = 0.251$ (25.1%; 1.12 ms high in each 4.46 ms period), rather than specifying a sinusoidal voltage amplitude. The selected point’s measured vector-RMS acceleration, 2.68 m/s^2 , was closest to the provisional 2.7 m/s^2 target. Appendix Figure B.1c, d shows both selections. The applied signal is a 0–9 V rectangular PWM waveform; the sine-shaped reference in panel (d) represents only its fundamental component at the drive frequency. These are mounted-system operating points, not perceptual constants [29].

3.1.2 Actuator Type

Driven at those settings the two types differed by more than an order of magnitude: across ten trials each, motion onset was $6.14 \pm 1.20 \text{ ms}$ for the LRA against $205.93 \pm 22.11 \text{ ms}$ for the ERM, and command-to-steady intervals $127.6 \pm 11.6 \text{ ms}$ and $491.5 \pm 24.9 \text{ ms}$ (Figure 3.2a, b). The ERM onset alone is roughly a quarter of the reaction times later observed and would enter measured latency as a condition-specific offset, so the LRA was adopted.

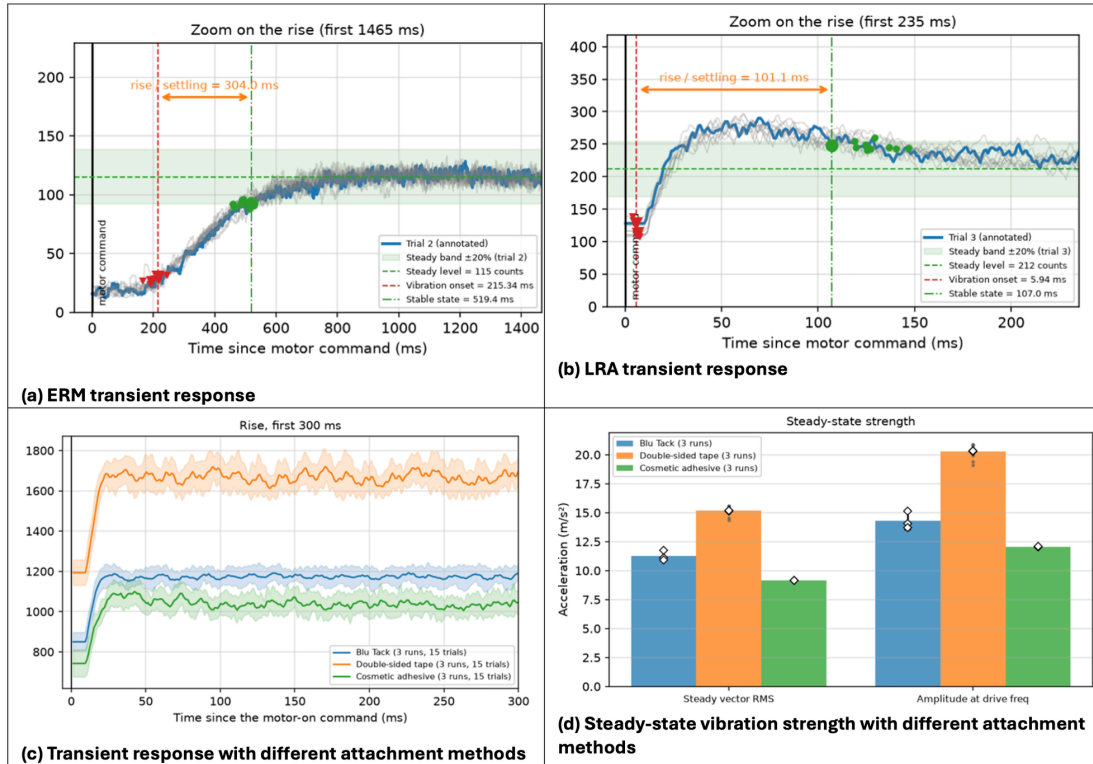


Figure 3.2: Transient response and attachment validation. (a, b) One annotated trial per actuator, drawn from ten trials each, showing the motor command, detected vibration onset, and the steady band used to score settling. (c, d) Rise and steady-state vibration strength for the three attachment methods (three mounts $\times$ five trials each).

3.1.3 Attachment Method

All three adhesives transmitted repeatable cues, with tape transmitting most and cosmetic adhesive least (Table 3.1; Figure 3.2c, d). Blu Tack was selected because it holds immediately, repositions without residue, and can serve a whole fitting. Actuator, attachment, and drive were fixed before recruitment.

Table 3.1: Steady-state vibration transfer by attachment method (three mounts $\times$ five trials); means of per-mount means, between-mount SD in parentheses.

Method	Mounts	Vector RMS (m/s ²)	Robust peak (m/s ²)	THD (%)	Relative (dB)
Blu Tack	3	11.27 (0.45)	17.24	23.9	0.00
Double-sided tape	3	15.18 (0.04)	23.99	11.6	+2.58
Cosmetic adhesive	3	9.14 (0.01)	15.81	17.2	-1.82

3.1.4 Stimulus Pool

The locked pool comprised nine sequences per difficulty level; all eleven components followed the intended ordering, adjacent-pair violations remained below the 10% limit, and every structural check passed (Appendix A).

3.2 Main User Study

The primary analysis compares matched visual (B) and vibrotactile (C) finger guidance; Condition A is retained only as an unguided descriptive reference.

3.2.1 Data Completeness

All 20 participants completed 540 trials and 16,200 recorded cue events. One carry-over was excluded, leaving 16,199 analysed events; manual review resolved every valid event. There were three timeouts in A, one in B, and none in C. Setting aside those four timeouts and the excluded carry-over leaves 16,195 keypresses, of which 4.50% had the automatic finger label overridden during that review (Appendix B, Figure B.2).

3.2.2 Condition Effects

Table 3.2 and Figure 3.3 show that C was faster and more finger-accurate than B; no participant reversed the reaction-time ordering.

Table 3.2: Condition summaries, group mean $\pm$ SD across the 20 participants. The primary contrast uses Main Finger Accuracy and complete-action reaction time. Condition A cues no finger, so its finger entries are agreement with the hidden generated fingering, not accuracy.

Measure	A (key only)	B (visual finger)	C (vibrotactile finger)
Key Accuracy (%)	99.1 $\pm$ 0.8	98.9 $\pm$ 1.4	99.7 $\pm$ 0.4
Main Finger Accuracy (%)	18.8 $\pm$ 2.8	94.7 $\pm$ 3.1	98.3 $\pm$ 1.2
Finger Accuracy correct key (%)	19.0 $\pm$ 2.8	95.7 $\pm$ 2.5	98.6 $\pm$ 1.2
RT, correct-key events (ms)	488.2 $\pm$ 55.2	970.2 $\pm$ 124.1	723.5 $\pm$ 91.9
RT, complete-action events (ms)	488.8 $\pm$ 66.6	969.9 $\pm$ 123.4	723.2 $\pm$ 91.2

Participant trajectories show ceiling-level key accuracy but clear separation of unguided A from B and C on finger-aware outcomes (Figure D.1).

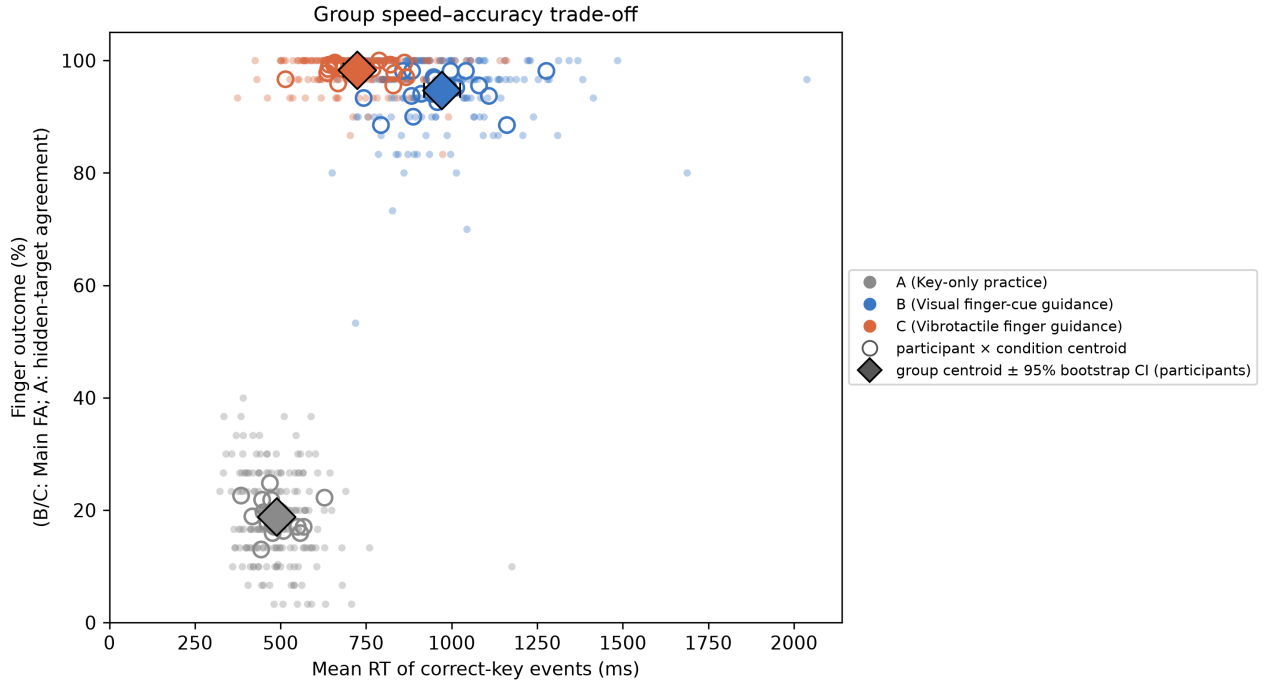


Figure 3.3: Speed–accuracy plane, reaction-time axis from zero. Small points are trials, rings participant $\times$ condition centroids, and diamonds group centroids with 95% participant-level bootstrap CIs ($n = 20$).

Main Finger Accuracy rose by 3.65 percentage points, 95% CI [2.39, 4.98], $t(19) = 5.35$, $p = 3.7 \times 10^{-5}$, $d_z = 1.20$, Wilcoxon $p = 0.00029$, with 19 of 20 participants higher under C. Reaction time fell by 247 ms, 95% CI [206, 287] ms, $t(19) = -11.54$, $p = 5.0 \times 10^{-10}$, $d_z = -2.58$, Wilcoxon $p = 1.9 \times 10^{-6}$, with all 20 faster under C. Key accuracy rose by 0.83 percentage points, 95% CI [0.31, 1.43], $t(19) = 2.82$, $p = 0.0109$, $d_z = 0.63$, remaining near ceiling under both conditions. Participant-level paired differences are shown in Figure 3.4. Condition A agreed with the never-shown fingering on 18.8% of events.

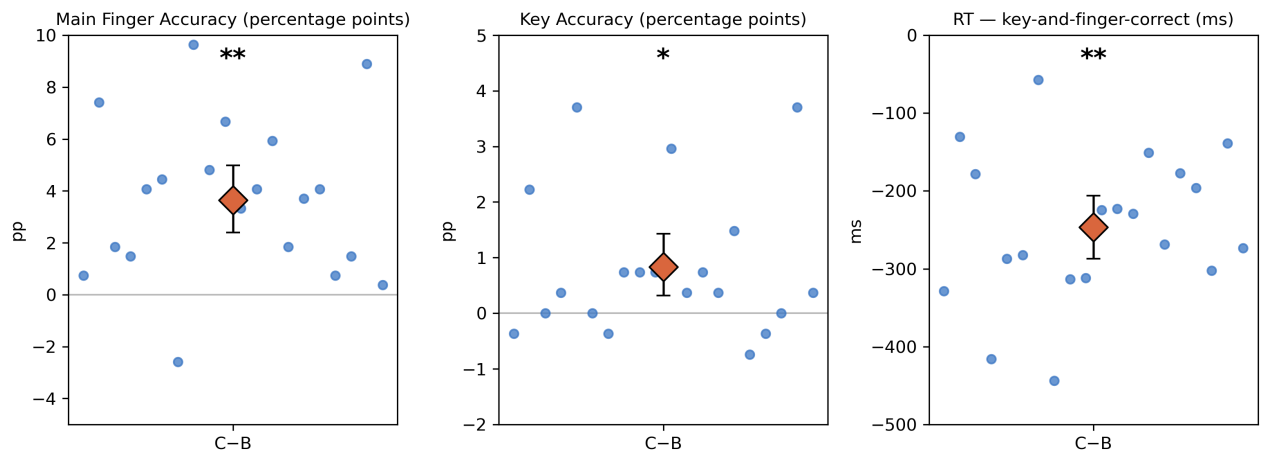


Figure 3.4: Paired C–B differences for the three principal outcomes. Points are participants; diamonds and bars are group means and 95% participant-level bootstrap CIs ($n = 20$). Negative reaction-time differences favour C. Stars show the corresponding two-sided paired t -tests: $*p < 0.05$, $**p < 0.01$.

3.2.3 Difficulty

Difficulty increased reaction time, $F(1.57, 29.80) = 31.61$, $p = 2.4 \times 10^{-7}$, $\eta_p^2 = 0.62$, while accuracy stayed within 94.2–95.4% under B and 98.2–98.4% under C. Its interaction with Condition was inconclusive, $F(1.69, 32.17) = 2.76$, $p = 0.086$, $\eta_p^2 = 0.13$. The B–C gap was 235, 212, and 293 ms across α , β , and γ ; the easy-to-hard change was +58 ms, 95% CI $[-16, +132]$ ms. Thus the data establish the stimulus ordering but neither dependence nor independence of the modality difference from load (Figure 3.5).

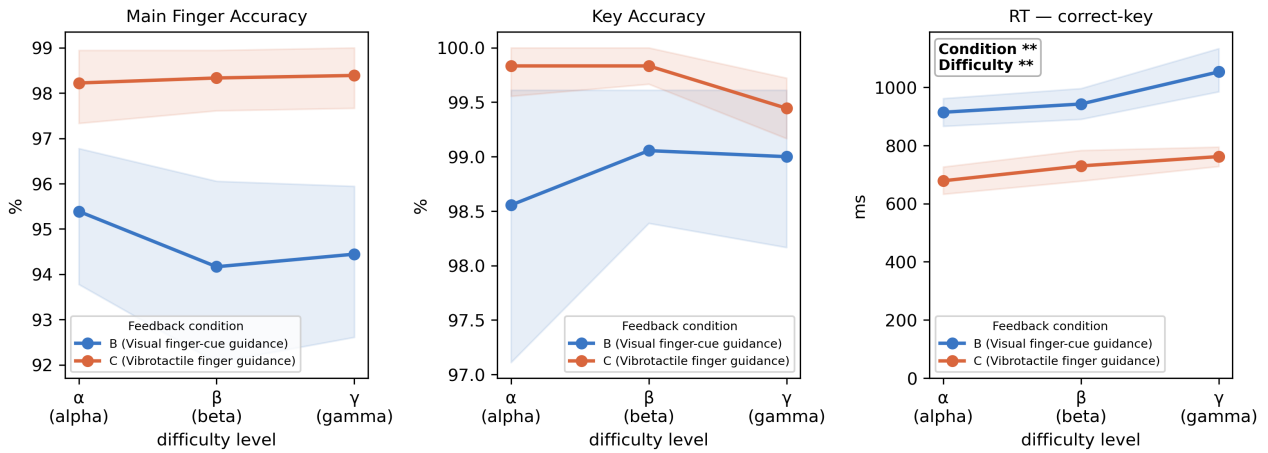


Figure 3.5: Outcomes by difficulty level and condition. Lines are group means and shaded bands are 95% participant-level bootstrap CIs ($n = 20$). One reaction-time panel is shown, on the correct-key definition; the complete-action definition differs from it by at most 1.3 ms in any cell. The accuracy panels are autoscaled because the data occupy a narrow 94–99% range. In the reaction-time panel, stars mark significant omnibus repeated-measures ANOVA effects: $*p < 0.05$, $**p < 0.01$; the Condition $\times$ Difficulty interaction was not significant.

3.2.4 Finger-Specific Effects

The reaction-time ANOVA returned effects of Condition, $F(1, 19) = 122.22$, $p = 1.0 \times 10^{-9}$, $\eta_p^2 = 0.87$, and Finger, $F(2.52, 47.94) = 26.18$, $p = 2.2 \times 10^{-9}$, $\eta_p^2 = 0.58$, plus a Condition $\times$ Finger interaction, $F(2.43, 46.25) = 7.35$, $p = 0.00087$, $\eta_p^2 = 0.28$. Vibrotactile reductions were largest for the middle and ring fingers (287 and 286 ms), 270 ms for the index, and smallest for the thumb and little finger (186 and 176 ms; Figure 3.6); every plotted 95% participant-level bootstrap CI excluded zero.

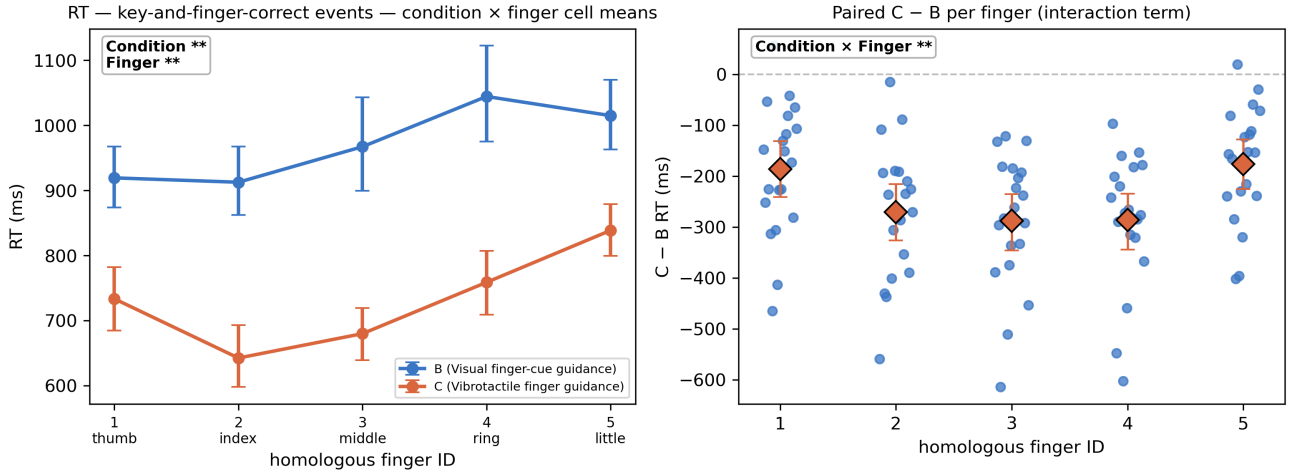


Figure 3.6: Reaction time by homologous finger. Left, condition $\times$ finger group means with 95% participant-level bootstrap CIs. Right, paired C–B differences per finger: points are participants; diamonds and bars are group means with the same CIs ($n = 20$). Stars mark the corresponding omnibus repeated-measures ANOVA effects: $*p < 0.05$, $**p < 0.01$.

Accuracy became more uniform under C, its per-digit spread falling from 3.9 to 1.7 points (Appendix Figure D.2), whereas reaction-time dispersion did not: the noise-corrected within-participant SD across fingers increased from 71 to 88 ms, and the coefficient of variation rose from 0.085 to 0.130, $t(19) = 4.63$, $p = 0.00018$ (Figure 3.7). The little finger combined a large accuracy gain (4.5 points) with the smallest time gain, distinguishing successful selection from the remaining cost of movement.

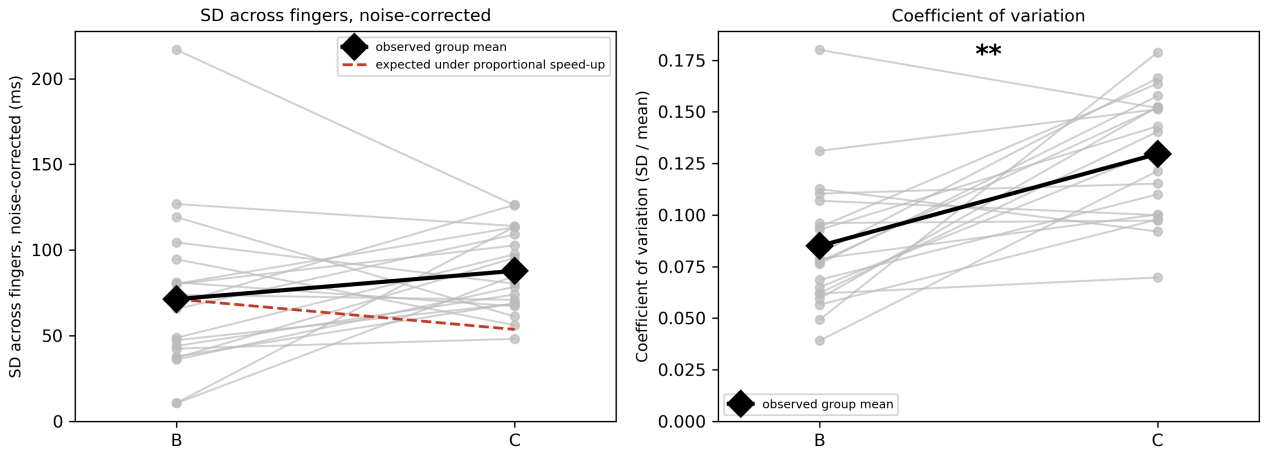


Figure 3.7: Reaction-time dispersion across fingers by condition. Grey lines pair participants and black diamonds and lines show observed group means ($n = 20$). Left, noise-corrected within-participant SD; right, coefficient of variation. The red dashed trajectory ends at the value expected under proportional speed-up and is not a confidence interval. Stars show the corresponding two-sided paired t -test ($*p < 0.05$, $**p < 0.01$).

Main Finger Accuracy was not entered into the ANOVA because it was ceiling-compressed, 66 of 200 participant $\times$ condition $\times$ finger cells sitting at exactly 1.0. A binomial mixed model on the same cells is not compromised by that ceiling and was fitted in its place (Figure 3.8). It gave an overall C-over-B odds ratio of 3.29, 95% CI [2.19, 4.94], $t(19) = 6.15$, $p = 6.5 \times 10^{-6}$, agreeing with the participant-level paired contrast, with a between-participant SD of 0.46 on the logit scale and per-digit odds ratios from 2.10 at the thumb to 6.92 at the middle finger. The Condition $\times$ Finger interaction the ANOVA could not supply was $\chi^2(4) = 10.90$, $p = 0.028$;

simulation of the same fitting procedure at $N = 20$ leaves that four-degree-of-freedom test mildly liberal, so it is read as suggestive rather than confirmatory. On the automatic pre-screen scores the C over B ordering held at every θ from 0.30 to 0.45 and converged at $\theta = 0.50$.

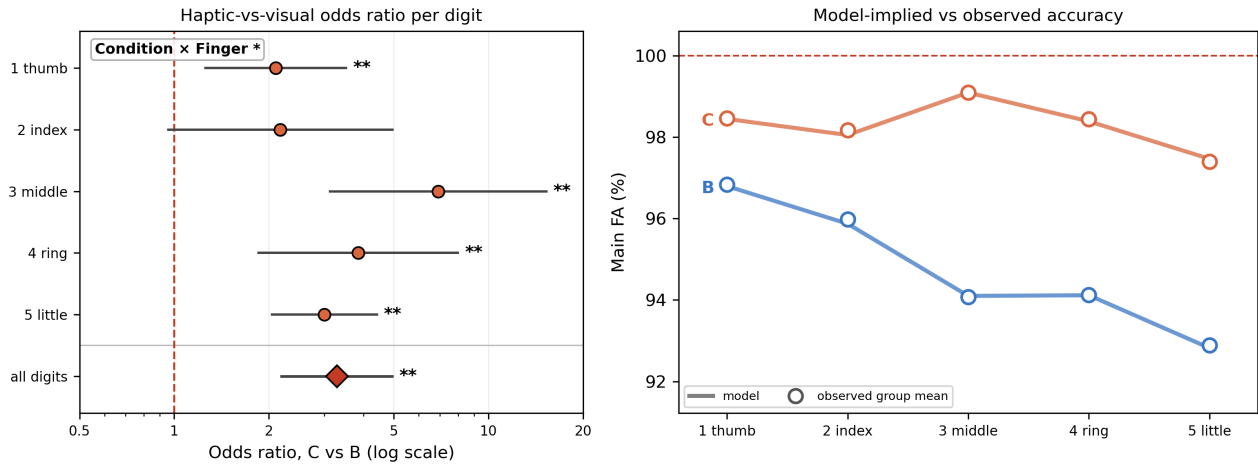


Figure 3.8: Binomial mixed model on the finger-accuracy cells, with a participant random intercept ($n = 20$). Left, the C–B contrast as an odds ratio within each digit, with 95% participant-clustered intervals; the diamond is the single overall effect from the additive model, the dashed line marks no difference, and the axis is logarithmic so equal ratios occupy equal space. Stars beside odds ratios use two-sided participant-clustered t -tests; the interaction annotation uses the likelihood-ratio test (* $p < 0.05$, ** $p < 0.01$). Per-digit tests are exploratory and unadjusted. Right, model-implied accuracy against the observed group means, the check that a ceiling-bounded fit reproduces the cells it was fitted to; the two lines are labelled B and C at their left-hand ends.

3.2.5 Structure of Finger Errors

Outcome composition separates correct-key events by finger agreement (Figure 3.9). Different-finger events dominated A but fell sharply under B and C; participant rates are in Figure D.3.

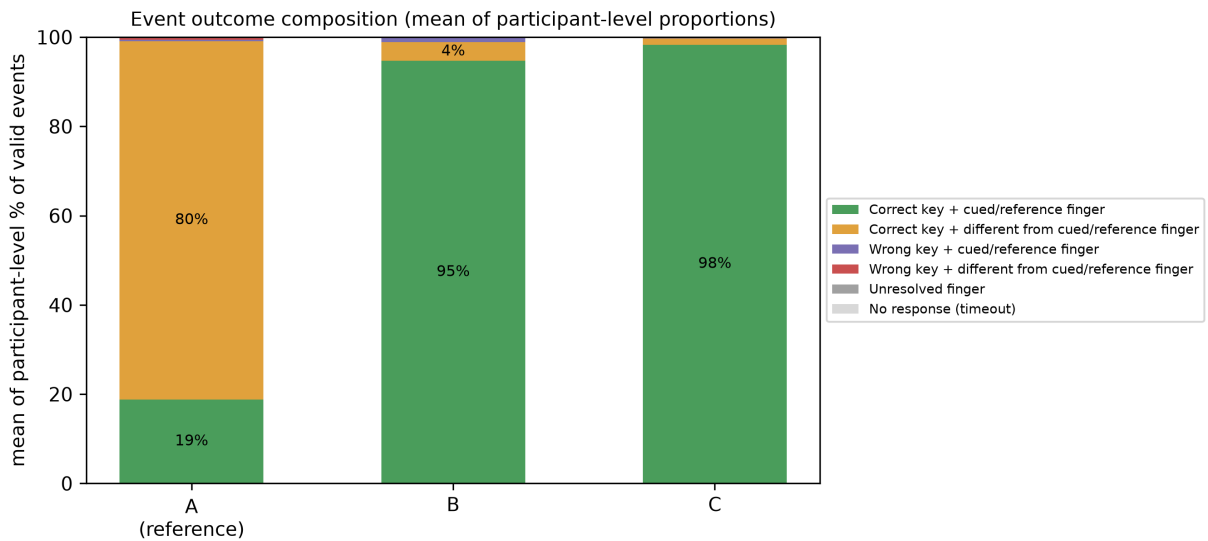


Figure 3.9: Event-outcome composition by condition, calculated within participant and then averaged ($n = 20$). The stacked categories distinguish key correctness, finger correctness, unresolved events, and timeouts.

Of the 10,799 cued events in Conditions B and C, 10,050 were assigned to the cued finger outright and 443 were off-diagonal events in which it still held at least θ of the mass — adjacent-fingertip ambiguity the rule forgave, not participant error. The remaining 306 are genuine substitutions, 233 under visual against 73 under vibrotactile guidance (Figure 3.10).

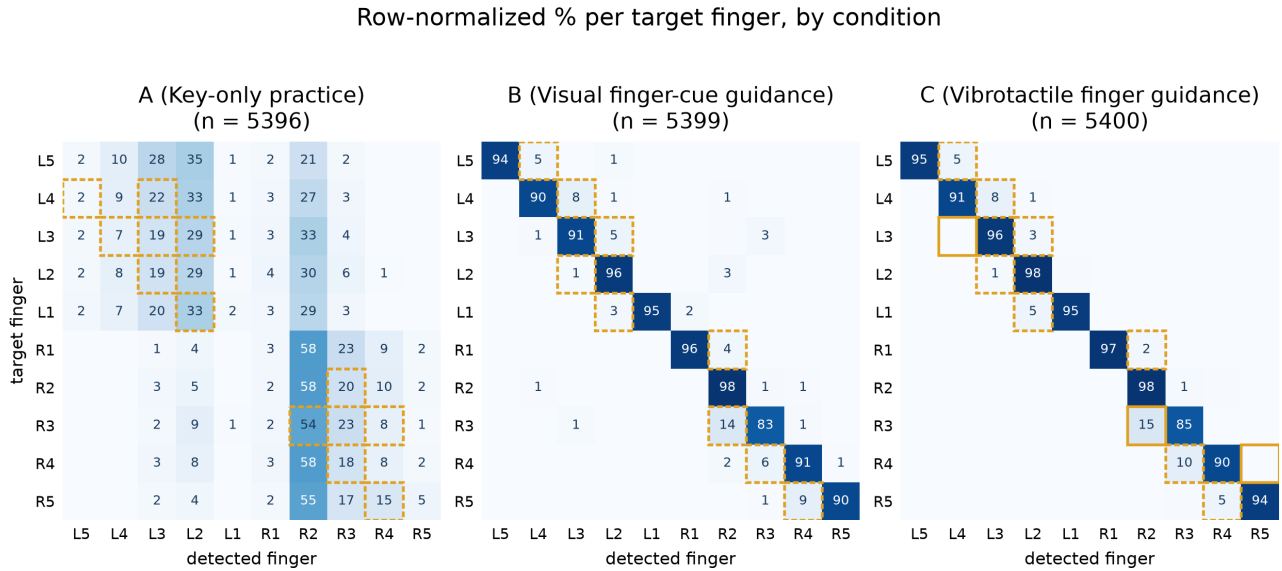


Figure 3.10: Target versus detected finger by condition, row-normalised, physical order L5–R5, pooled over 16,195 resolved responses from 20 participants; the four timeout events and the one excluded carry-over are absent from the matrix. Condition A cues no finger, so its rows describe unguided choice, not accuracy.

Under visual guidance, 67 of 233 substitutions (28.8%) used the wrong hand; under vibrotactile guidance, 2 of 73 (2.7%). Because both conditions contained 5,400 planned events, their counts are directly comparable. Same-hand substitutions fell from 166 to 71, but cross-hand substitutions from 67 to 2.

Within-hand errors were mainly single-position neighbour slips in both conditions (80.1% and 73.2%). Of the visual cross-hand substitutions, 53 (79%) used the homologous digit, against 20% under uniform within-hand choice; the same number were left-hand targets executed with the right hand, most often R2. These pooled counts describe error composition rather than a participant-level test.

The exploratory model estimated stronger hand evidence under C but similar within-hand digit sharpness. Held-out risk enrichment did not exceed a generic multinomial benchmark; full specification and checks are in Appendix C.1.

3.2.6 Change Within the Session

Reaction time improved with repetition, $F(1.72, 32.65) = 58.22, p = 7.2 \times 10^{-11}, \eta_p^2 = 0.75$, while Main Finger Accuracy moved little. The Condition $\times$ Repetition interaction was small, $F(1.79, 33.96) = 0.13, p = 0.853, \eta_p^2 = 0.01$: the B–C gap changed from 253 ms at the first repetition to 239 ms at the third, a change of -14 ms, 95% CI $[-70, +43]$ ms (Figure 3.11; difficulty-aligned all-condition trajectories are in Appendix Figure D.4).

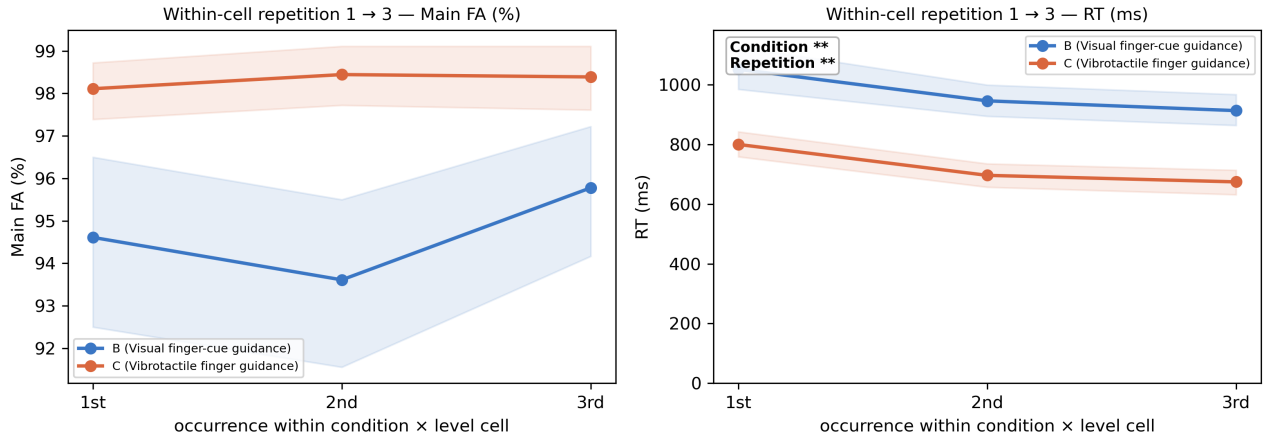


Figure 3.11: Main Finger Accuracy and reaction time across the three repetitions of each condition × difficulty cell. Lines are group means and shaded bands are 95% participant-level bootstrap CIs ($n = 20$); the reaction-time axis begins at zero. In the reaction-time panel, stars mark significant omnibus repeated-measures ANOVA effects ($*p < 0.05$, $**p < 0.01$); the Condition × Repetition interaction was not significant.

Appendix Figure D.4 complements the controlled Condition × Repetition analysis by aligning, within each participant and difficulty, the nine trials by actual session order. From the first to ninth occurrence, observed group-mean reaction time fell from 733 to 581 ms for α , 901 to 613 ms for β , and 989 to 743 ms for γ , while key accuracy stayed between 98.0 and 100.0%. Because the randomised condition mix differs across occurrences, these raw trajectories are descriptive rather than estimates of a practice effect; condition-adjusted values remain available in the tidy export.

3.2.7 Unguided Finger Choice in Condition A

Condition A describes free choice, not modality performance. Four digits absorbed 85.2% of presses, R2 alone 42.5%, and participants used an effective 4.81 ± 1.30 of ten fingers across Condition A (Figure 3.12).

Condition A — free-fingering preference and strategy

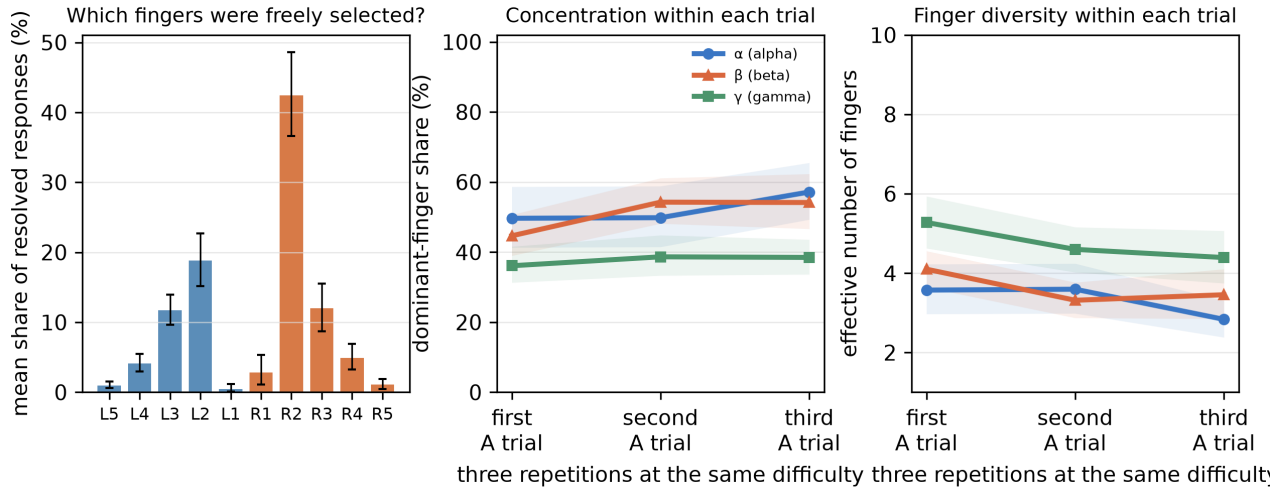


Figure 3.12: Unguided finger choice in Condition A. Left, share of resolved presses per digit; middle and right, concentration on the dominant digit and effective number of digits across repetitions and difficulty levels. Bars and lines are group means; error bars and shaded bands are 95% participant-level bootstrap CIs ($n = 20$).

Right-hand use exceeded the near-balanced generated fingering by 13.8 percentage points, $t(19) = 6.55$, $p = 2.8 \times 10^{-6}$; the excess was concentrated in right-handed participants (Appendix Figure D.5).

Same-hand finger-transition distance was roughly half the generated value at every level, while key travel did not change consistently (Figure 3.13). Yet key accuracy remained 99.1% while fingering agreement was 18.8%, showing that MIDI-only scoring misses the behaviour finger guidance is intended to change.

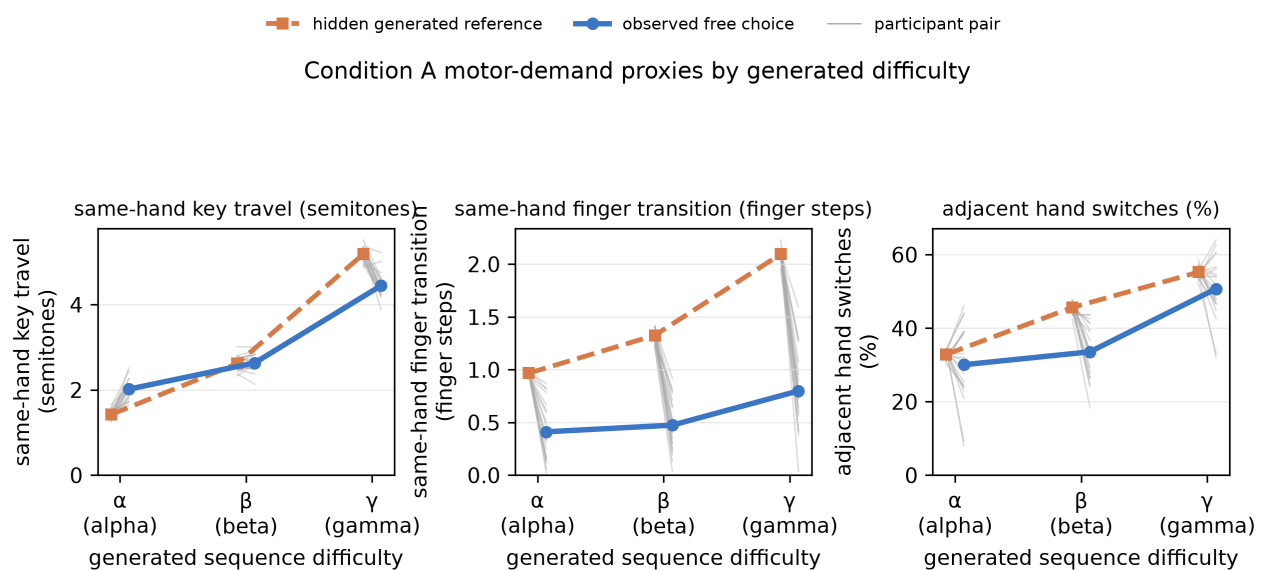


Figure 3.13: Condition A motor-demand proxies by generated difficulty. Dashed lines show the hidden generated reference, solid lines observed free choice, and thin lines participant pairs ($n = 20$).

3.3 Tele-training Deployment

All 1,000 transport probes arrived, with median round-trip time 36.6 ms; 35.5% exceeded 100 ms. Across 244 learner-side cues, dispatch took a median 1.1 ms; among 160 dual-modality cues, visual–haptic software skew stayed within 0.47 ms. The measured components imply about 55 ms from note-on to cue typically and 170 ms at p95. Thus the deployment delivered both modalities with tight local alignment, but network tail latency remains relevant to synchronous performance. Single-operator demonstrations verified the complete path but did not estimate a remote behavioural effect (Appendices [B](#) and [D](#)).

Chapter 4

Discussion

4.1 What the Experiment Establishes

RQ1 is answered: with target-finger information and sequence difficulty matched, vibrotactile cues reduced reaction time by 247 ms and increased finger accuracy by 3.65 percentage points relative to visual cues. **RQ2 is answered:** the difference varied across digits and largely removed the wrong-hand error class admitted by visual guidance. These patterns constrain the mechanism more strongly than a mean latency alone.

The methodological contribution is a ten-finger nail-mounted array that leaves the fingerpad free, a reviewed camera–MIDI measurement chain that records which digit acted, and a generator whose difficulty ordering was validated before use. Substantively, the results show that equal information delivered symbolically and somatotopically is not interchangeable: modality changed both action time and the identity of the selected effector.

4.2 Relation to Vibrotactile Guidance of the Hand

Coscia and Al Borno found lower timing error with vibrotactile than visual stimulation in novice piano learning, with no significant note-accuracy difference [10]. Their conditions, however, differed in both channel and content: vibration specified the finger, whereas the visual condition flashed the key without identifying a finger. Here both conditions named the same digit and shared the same target-key light, isolating the delivery frame. Key accuracy remained near ceiling, while the modality difference appeared in finger selection. The paradigms are nevertheless complementary: their outcome was measured after cue withdrawal, whereas this study measured concurrent guidance. Their note-accuracy point estimate also favoured visual stimulation despite not reaching significance. The combined evidence therefore cautions that concurrent and retained advantages may dissociate; an immediate advantage cannot be assumed to imply retention.

Hapticus compared vibration, electrical muscle stimulation, and a linkage exoskeleton and found no modality dominant across outcomes [11]. Those devices differ in physical assistance as well as sensory channel: a linkage partly imposes movement, whereas vibration signals without moving the digit. The present comparison stays at that informational end. The combined evidence suggests a more precise design rule than “haptics helps”: stimulation site naturally encodes a categorical, body-addressed variable such as *which finger*, but is less suited to continuous quantities such as movement magnitude. Hardware still matters because its dynamics must be short relative to the effect being measured, which motivated the LRA selection before recruitment.

4.3 Where the Benefit Falls Across the Hand

The Condition $\times$ Finger interaction makes the pattern more diagnostic than the overall effect. Lang and Schieber separated passive mechanical coupling from active neuromuscular limits [2]: the thumb was most independent, the index, middle, and ring fingers least mechanically independent, and the little finger intermediate. Active control produced a different grouping, constraining the ring and little fingers most. Reaction-time reductions followed the mechanical grouping, being largest for the index, middle, and ring fingers and smallest for the thumb and little finger. The little finger is especially informative because it is only intermediately coupled yet actively constrained: a cue can resolve its identity without removing the control cost of producing its movement. This is a post hoc rank correspondence rather than a tested association, but it is informative because the literature-derived ordering did not enter the analysis.

The little finger distinguishes selection from execution. Vibrotactile guidance produced one of its largest accuracy gains but its smallest time gain, suggesting that the cue resolved which digit should act without removing the cost of mobilising that digit. Conversely, index accuracy was already near ceiling, leaving its benefit to appear mainly in time. The tactile condition narrowed the per-digit accuracy spread while widening the time spread, consistent with removal of a cue-dependent selection cost while digit-specific movement costs remain. Furuya and Altenmüller group the index differently [3], so the present correspondence is not unique evidence for mechanical coupling. It predicts, rather than demonstrates, a smaller modulation in trained pianists whose individuation is more equalised by practice.

4.4 The Structure of the Errors

Wrong-hand actions accounted for 28.8% of visual substitutions but 2.7% of vibrotactile substitutions. This reduction was selective: within-hand substitutions fell by a factor of 2.3, cross-hand substitutions by a factor of 34. A general increase in cue salience should reduce error classes more proportionally; the observed selectivity falls on the dimension directly supplied by a body-located cue.

Moreover, 79% of visual wrong-hand actions preserved the homologous digit, compared with 20% expected under uniform substitution within that hand. Such an error retains finger number but loses hand assignment, consistent with a failure in mapping a symbolic label to the body. Attention or action-planning errors could produce the same pattern, so it is not a unique signature of symbolic mapping. A site-delivered cue nevertheless makes hand membership part of cue location, although the two tactile wrong-hand errors show that it does not make misassignment impossible. Neighbour slips remained common within each hand and are more compatible with ambiguous or imprecise execution than with failure of hand mapping.

Visual wrong-hand errors were usually left-hand targets executed with the right hand, often R2, which also dominated free choice in Condition A. Habitual choice therefore appears as a residual mode under symbolic guidance and is suppressed by somatotopic guidance. The pooled error counts are too sparse under C for participant-level inference, so this is a bounded compositional argument. The exploratory model generalised risk to held-out participants but did not outperform a multinomial benchmark; its value is interpretable decomposition, not proof of a unique mechanism (Appendix C.1).

4.5 Competing Explanations

The preferred account is that somatotopic delivery removes an external-to-body transformation required by the display, but the data do not identify a single mechanism. The selective error reduction disfavours a wholly non-specific salience account, as does the 1.6-fold variation in time benefit across digits. Symbolic-to-body translation and modality-specific salience remain possible contributors, and the present design cannot apportion them.

Perceptual onset was also unmatched. Dispatch timestamps precede both display appearance and actuator motion; the LRA begins moving about 6.1 ms after command and reaches its steady band later, while display latency was not measured. The mechanical settling interval cannot be treated as a known tactile disadvantage because perceptual detection may occur anywhere on the ramp. Nor can the residual be assumed small solely because both pathways contain delay: their unmeasured components may differ in either direction. This uncertainty affects the magnitude of the reaction-time contrast but not the accuracy or error-composition results. Direct detection-latency measurements are needed to resolve its sign and size.

4.6 What the Two Null Interactions Do and Do Not License

Difficulty strongly increased reaction time, but the Condition $\times$ Difficulty interaction was inconclusive ($p = 0.086$, $\eta_p^2 = 0.13$). The gap changed by +58 ms from the easiest to hardest level, 95% CI $[-16, +132]$ ms. This interval admits both little change and substantial widening; a moderate interaction is plausible but unresolved at $n = 20$. The data therefore establish neither dependence on nor independence from load.

The repetition result is more constraining. Practice was large ($\eta_p^2 = 0.75$), yet the between-condition gap changed by only -14 ms, 95% CI $[-70, +43]$ ms, over three repetitions. The interval excludes rapid convergence on the scale of the 247 ms mean gap, providing bounded evidence against the visual disadvantage being only initial unfamiliarity with the display. It cannot address automatisisation over days or weeks (Figure 3.11).

4.7 Unguided Finger Choice and What Guidance Must Overcome

Condition A required only the correct key, whereas B and C also constrained the finger; it is therefore descriptive, not a performance baseline. Free choices concentrated on four digits, favoured the right hand, and shortened finger transitions without consistently shortening key travel. Most importantly, 99.1% key accuracy coexisted with only 18.8% agreement with the hidden fingering. A MIDI-only evaluation would therefore report near-perfect performance while missing the behaviour the system is intended to shape (Figures 3.12 and 3.13; Appendix Figure D.5).

4.8 Guided Performance Is Not Learning

All behavioural outcomes were measured while the cue was present. Augmented feedback can improve acquisition performance yet impair retention when learners depend on it; Salmoni and colleagues therefore distinguish guided performance from learning using a transfer test without feedback [6]. Sigrist and colleagues describe the same guidance hypothesis and note that concurrent feedback can improve acquisition without improving retention, although results vary with task complexity [7]. The present 247 ms advantage is consequently not evidence of better learning and could coexist with poorer retention.

The specificity-of-learning hypothesis gives an opposing prediction [7]. Learners may incorporate the most useful afferent source into the task itself. A visual diagram has no counterpart after withdrawal, whereas a somatotopic cue is delivered in the body’s coordinate frame. Removing tactile input may therefore remove a useful signal without removing the frame in which the movement was practised. Guidance predicts a larger withdrawal cost after tactile cueing; specificity predicts that practising in the body-referenced frame may transfer better than practising from an external diagram. A withdrawal phase and delayed retention test are required to distinguish these accounts. Until then, the claim is limited to immediate cue-guided performance.

4.9 Limitations

Sample and duration. Twenty participants from one institution, mostly without keyboard experience and with a 19:1 sex imbalance, limit generalisation of effect magnitude. Hundreds of events improve measurement within each participant but do not increase the number of independent participants. A single session without withdrawal cannot establish retention or transfer; sex differences are not interpretable from one woman.

Finger labels. Every scored event was visually reviewed, but one reviewer provides no inter-rater reliability estimate. Systematic rater bias would affect finger accuracy and error composition, though not MIDI-derived reaction time. The threshold rule also forgives near-ties when the target retains sufficient probability mass. Automatic-score sensitivity checks preserved the C-over-B ordering from $\theta = 0.30$ to 0.45; at 0.50 the means converged, and absolute accuracy remains conditional on the chosen $\theta = 0.40$.

Ceiling and multiplicity. Main Finger Accuracy equalled 1.0 in 66 of 200 participant $\times$ condition $\times$ finger cells, preventing a suitable accuracy ANOVA and making per-digit differences mainly ordinal. That argument bars the F ratio rather than every model, so the same cells were also fitted as a binomial mixed model with a participant random intercept, which a ceiling does not compromise; it agreed with the participant-level contrast in direction and significance, so the accuracy conclusion does not depend on aggregating events to participant means. Per-digit comparison remains weaker than that, because the ranking by percentage points and the ranking by odds ratio exchange the ring and little fingers: the same change in odds is fewer points at a lower baseline, and neither scale is the self-evidently correct one on which to compare digits. Ceiling compression may hide further separation but does not justify assuming its size or direction. Only the per-digit reaction-time model was pre-specified; difficulty and repetition models used the same data without multiplicity correction. Neither secondary interaction supports a confirmed null.

Task reduction. Single-key events, muted audio, white keys, and self-paced responses isolate cue interpretation but exclude polyphony, expressive timing, and external tempo [27, 28]. Condition A represents self-selected strategy among mostly inexperienced participants rather than expert fingering.

Physical timing and tele-training. Cue dispatch did not measure physical visual and tactile onset, so the reaction-time effect cannot be corrected to perceptual onset. In the remote deployment, local software alignment was tight but transport was variable, and the single-operator demonstrations lacked counterbalancing and manual review. They establish delivery, not a remote behavioural effect; more than a third of probes exceeded 100 ms, which would matter whenever teacher and learner must share a beat.

4.10 Future Work

Three studies follow in priority order. First, test cue withdrawal and delayed retention across multiple sessions, powering the study on participants rather than events. Second, use onset matching, a hand-congruent visual cue,

expert participants, and an independent label rater to separate symbolic mapping, individuation, and measurement. Third, repeat the counterbalanced, manually reviewed remote comparison while controlling network tail latency. Later work may add external tempo or other finger-assignment tasks.

Chapter 5

Conclusion

The project aim was met within its stated scope: a nail-mounted platform delivered and measured finger-specific guidance without occupying the fingerpad, supported a working tele-training deployment, and enabled a matched comparison with visual cues. In that comparison, vibrotactile guidance reduced reaction time by 247 ms and improved finger accuracy by 3.65 percentage points. Digit-dependent benefits and the marked reduction in cross-hand substitutions indicate an effect on selecting which finger should act. These findings answer RQ1 and RQ2 for immediate cue-guided performance in one session; they do not establish retention after cue withdrawal or behavioural benefit during remote use.

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Appendix A

Stimulus Pool Provenance and Validation

The formal stimulus pool was generated and locked with

```
generation_seed = 1799551287,
```

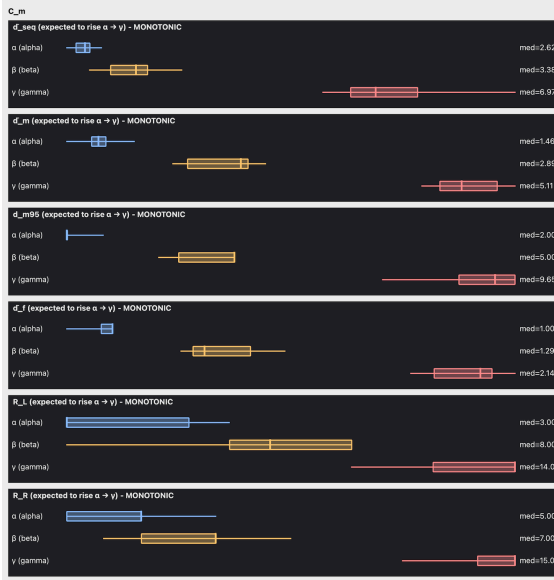
nine sequences per difficulty bin, on the calibrated profile spanning MIDI notes 48–72. Saved sequence names carry the seed as their batch prefix (e.g. 1799551287- α -1). Rerunning the generator with the same profile, note range, family size, and seed at the code state recorded in Table A.1 reproduces the pool exactly, so the 27 complete 30-event action lists are not reprinted here: the stored pool itself, not a printed transcript, is what a replication should be checked against. Table A.1 gives that provenance, and Table A.2 summarises the per-level validation statistics behind the **VALID** conclusion visualised in Figure A.1.

Table A.1: Provenance of the locked stimulus pool: the seed, code state, and stored locations from which the 27 sequences can be regenerated and checked.

Item	Value
Generation seed	1799551287
Pool	3 difficulty levels $\times$ 9 sequences = 27
Sequence length	30 cue events, one complete action each
Note range	MIDI 48–72, white keys only (15-note pool)
Generator module	main/app/sequence_generator.py
Validation module	main/app/stimulus_validation.py
Generator last changed	commit 6d4d2c4 (2026-08-03)
Locked stimuli	main/data/sequence/1799551287- $\{\alpha, \beta, \gamma\}$ - $\{1..9\}$ /
Validation record	main/data/sequence_validation/1799551287/

Table A.2: Per-level validation summary of the locked pools: median [Q1, Q3] per component ($n = 9$ per level), adjacent-pair violation rate (limit 10%), and verdict. Distances are in semitones on the calibrated profile.

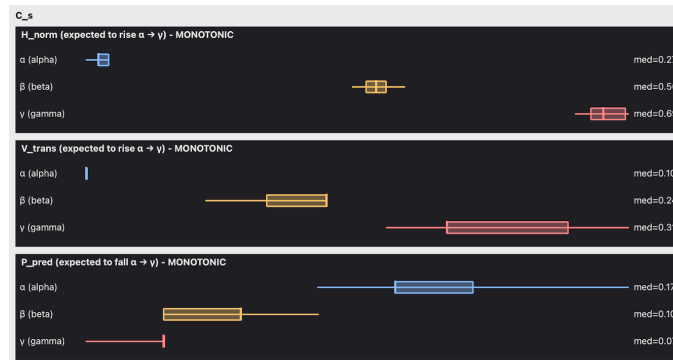
Component	Expected	α	β	γ	Viol.	Verdict
$d_{\text{seq}}^{\text{seq}}$	rise	2.62 [2.48, 2.69]	3.38 [3.00, 3.55]	6.97 [6.59, 7.59]	1.2%	monotonic
d_{m}^{m}	rise	1.46 [1.39, 1.54]	2.89 [2.36, 2.96]	5.11 [4.89, 5.46]	0.0%	monotonic
d_{95}^{m}	rise	2.00 [2.00, 2.00]	5.00 [4.00, 5.00]	9.65 [9.00, 10.0]	0.0%	monotonic
d_{f}^{f}	rise	1.00 [0.96, 1.00]	1.29 [1.25, 1.43]	2.14 [2.00, 2.18]	0.0%	monotonic
R_L	rise	3.00 [3.00, 6.00]	8.00 [7.00, 10.0]	14.0 [12.0, 14.0]	3.7%	monotonic
R_R	rise	5.00 [3.00, 5.00]	7.00 [5.00, 7.00]	15.0 [14.0, 15.0]	9.9%	monotonic
H_{norm}	rise	0.27 [0.27, 0.28]	0.51 [0.50, 0.51]	0.69 [0.68, 0.71]	0.0%	monotonic
V_{trans}	rise	0.10 [0.10, 0.10]	0.24 [0.21, 0.24]	0.31 [0.31, 0.38]	0.0%	monotonic
P_{pred}	fall	0.17 [0.17, 0.21]	0.10 [0.07, 0.10]	0.07 [0.07, 0.07]	0.0%	monotonic
A_h	rise	0.34 [0.31, 0.34]	0.45 [0.45, 0.48]	0.55 [0.55, 0.59]	0.0%	monotonic
H_{hand}	diagnostic	0.95 [0.94, 0.95]	0.99 [0.99, 0.99]	0.99 [0.98, 1.00]	—	reported only
B_h	matching	0.87 [0.87, 0.87]	0.93 [0.93, 0.93]	0.93 [0.93, 0.93]	—	above floors
O_{LR}	rise	0.00 [0.00, 0.00]	0.07 [0.06, 0.08]	0.26 [0.25, 0.27]	0.0%	monotonic
X_f	cross limits	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.13 [0.10, 0.20]	—	within limits
X_e	cross limits	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	3.00 [3.00, 3.00]	—	within limits



(a) Motor movement cost C_m .



(b) Bimanual coordination C_c .



(c) Sequence complexity C_s .

Figure A.1: Per-level distributions of every difficulty component over the locked pools ($n = 9$ sequences per level). Table A.2 reports medians, IQRs, violation rates, and verdicts.

All structural checks pass: every locked sequence uses both hands and satisfies the rejection rules (no repeated

three-event chunk, no hand above the 60% share, no finger run longer than three, consistent fingering on repeated keys).

Appendix B

Generator, Synchronisation, and Matching Detail

This appendix records the implementation detail omitted from Chapter 2: the bench-validation rig and drive sweeps, the stimulus generator’s internals and difficulty components, the video–MIDI synchronisation self-checks, the calibration of the finger-matching decision rule, and the tele-training measurement protocol. The Python entry points are:

- generation: `main/app/sequence_generator.py`;
- validation: `main/app/stimulus_validation.py`;
- anchoring: `main/app/sync_led.py`; and
- matching: `main/app/finger_matching.py`.

This appendix and that code are the reproduction reference for the locked pools of Appendix A.

B.1 Bench Validation Rig and Drive Sweeps

All three bench experiments of Section 2.2 drove the actuator through the experimental firmware and scored vibration from a coupled LIS3DH streaming three axes at 1.344 kHz over a shared SPI bus, intensity being the demeaned three-axis vector RMS over N samples,

$$a_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{i=1}^N [(x_i - \bar{x})^2 + (y_i - \bar{y})^2 + (z_i - \bar{z})^2]}.$$

Per-axis demeaning removes gravity, static offset, and mounting orientation together; a $\sqrt{x^2 + y^2 + z^2}$ magnitude is only second-order sensitive to vibration perpendicular to gravity and so mixes rig pose into every reported intensity.

Drive parameters. Frequency was swept from 100–350 Hz and re-swept at 1 Hz resolution around the peak, locating the mounted resonance at 224 Hz. The firmware’s PWM amplitude parameter, amp_{PWM} , was then swept from 4 to 128 at that resonance. It is an 8-bit setting mapped to duty cycle as $D = \text{amp}_{\text{PWM}}/255$; the driver therefore applies a rectangular 0–9 V pulse train, with high time D/f , rather than a sinusoidal voltage. The dashed $\sin(\pi D)$ curve in Figure B.1d is only a scaled model of the PWM fundamental at the drive frequency, not the applied waveform. The selected 25.1%-duty point produced 2.68 m/s^2 , the measured value closest to the provisional target of 2.7 m/s^2 within the $2.4\text{--}3.0 \text{ m/s}^2$ band; it was not the strongest setting. The band rescales the drive range this project had previously used onto the demeaned vector-RMS metric. It is not an independent psy-

chophysical calibration, so the selection fixes a reproducible operating point rather than a perceptually justified intensity.

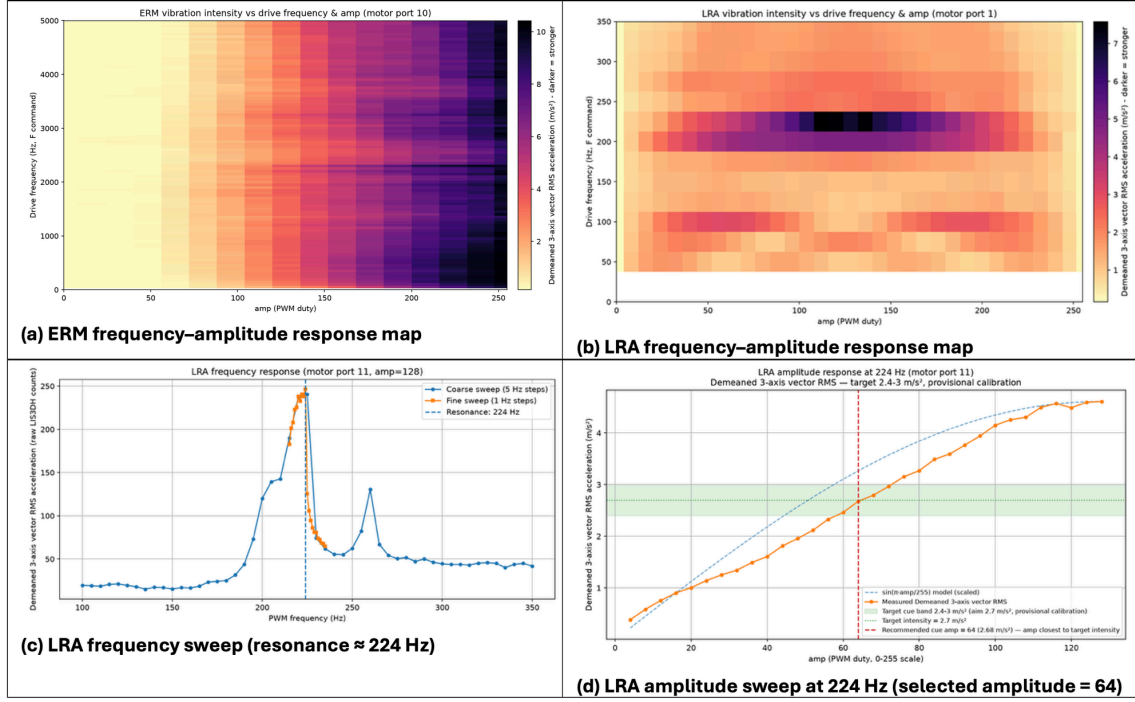


Figure B.1: Actuator characterisation and LRA parameter selection, one mounted assembly per actuator. (a, b) Vibration intensity over the frequency–amplitude plane for the ERM and LRA. (c) LRA frequency sweep locating the mounted resonance. (d) PWM–amplitude sweep at 224 Hz with the target cue band and selected PWM amplitude parameter.

Actuator comparison. Actuator and accelerometer were rigidly coupled and suspended freely, since rigid bench mounting damps the calibrated cue below reliable onset detection. Each actuator ran under its own study configuration — the LRA at its calibrated 224 Hz resonance, and the ERM on a 1 kHz PWM carrier at 19.6% duty because a DC motor cannot start under low-frequency chopping. Onset was estimated by CUSUM change-point detection [30], which does not require a fixed amplitude criterion, and settling time was scored separately as the interval from detected onset to entry into the steady band, so that when the actuator began to move and when it reached working amplitude are reported as distinct quantities.

Attachment. Each adhesive was characterised over three independent mounts of five trials at the fixed drive setting, with the mount as the averaging unit, since re-applying an adhesive is itself a source of variation. Reported quantities are the mean of per-mount means, the between-mount standard deviation, a robust peak, and total harmonic distortion (Table 3.1).

B.2 Difficulty Components

Motor cost C_m collects, in semitones, the mean displacement between consecutive events $\overline{d^{\text{seq}}}$, the mean and 95th-percentile same-hand key displacement $\overline{d^m}$ and d_{95}^m , the mean finger-transition distance $\overline{d^f}$, and the per-hand range R_h . Sequence complexity C_s classifies each transition by hand switch, binned displacement, and binned finger jump, and summarises the resulting distribution by normalised class entropy H_{norm} over $K = 16$ classes, class variety V_{trans} , and predictability $P_{\text{pred}} = \max_{\delta} \hat{p}(\delta)$. The last is estimated over *relative* moves

$\delta_t = (I(h_{t+1} \neq h_t), f_{t+1} - f_t, m_{t+1} - m_t)$ rather than over absolute actions, since a count-based estimate on 30 events would give almost every once-observed action probability 1 and so score the most varied sequences as the most predictable. Coordination complexity C_c comprises hand alternation A_h , hand balance B_h , the overlap O_{LR} of the hands' used note intervals, and the frequency X_f and extent X_e of cross-region events.

B.3 Hand Regions and Cross-Region Rules

Given the effective note bounds $k_{\min}$, $k_{\max}$, span $S = k_{\max} - k_{\min}$, and midpoint $k_{\text{mid}} = k_{\min} + S/2$, each level defines hand operating regions from an overlap ratio r :

$$R_L(r) = \left[k_{\min}, k_{\text{mid}} + \frac{rS}{2} \right], \quad R_R(r) = \left[k_{\text{mid}} - \frac{rS}{2}, k_{\max} \right],$$

with $r = 0$ for α (strict split, no crossing), $r = 0.15$ for β (small shared middle band), and $r = 0.30$ for γ . When the experimenter narrows the note range, $k_{\min}$, $k_{\max}$, and S follow the selected range, so hand regions, midpoint, and every span-normalised constraint describe the keyboard area actually in use; the effective bounds are stored in each saved sequence's metadata so that later recomputation uses the same span as generation.

A cross-region event occurs when the left hand is assigned above k_{mid} or the right hand below it:

$$X_t = I((h_t = L \wedge m_t > k_{\text{mid}}) \vee (h_t = R \wedge m_t < k_{\text{mid}})),$$

with frequency $X_f = \frac{1}{T} \sum_t X_t$ and extent $X_e = \max_t X_t |m_t - k_{\text{mid}}|$. Only γ permits $X_t = 1$, and then only when all ergonomic checks pass: $X_e/S \leq 0.20$; no more than two consecutive cross-region events; no jump into or out of a crossing greater than $0.45S$; and, in a crossed adjacent opposite-hand pair (the left-hand note lying above the right-hand note), the hand on the wrong side of the midpoint must use a thumb-side finger (1–3), since hand-over crossing with the ring or little finger is not a plausible movement.

Two further coordination components are defined as follows. Hand alternation is $A_h = \frac{1}{T-1} \sum_{t=2}^T I(h_t \neq h_{t-1})$. The hands' used note intervals are $U_h = [\min_{t \in Q_h} m_t, \max_{t \in Q_h} m_t]$ over each hand's active indices Q_h , and their overlap is $O_{LR} = |U_L \cap U_R| / |U_L \cup U_R|$, with interval sizes counted as the number of semitone positions covered inclusive of both endpoints. That convention is what makes the β -level combination of $X_f = 0$ with $O_{LR} > 0$ satisfiable: both hands must reach the midpoint note, but neither may enter the opposite register.

B.4 Candidate Construction and Rejection Rules

The generator samples a target level, then a left/right hand-label sequence satisfying that level's A_h and B_h constraints, then enumerates legal continuations satisfying the same-hand movement constraints, the hand-region rule, the overlap and cross-region constraints, the requirement that both hands appear, and the rule that each event contains exactly one action. If no legal continuation exists the candidate is discarded and generation restarts.

Legal continuations are sampled with level-specific proposal heuristics — persistence toward the dominant transition class and the dominant complete relative move, note locality around hand switches, and level-typical finger-jump windows — because several constrained bands, notably the α -level H_{norm} range, are reached with negligible probability by uniform sampling over 29 transitions. These heuristics shape only the proposal distribution: acceptance of a finished candidate is decided solely by the constraint table and the rejection rules, so no sequence outside the intended level ranges can be admitted.

A completed candidate is rejected if any level-defining component of C_m , C_s , or C_c lies outside the target

level’s range, if any event contains more than one action, if it contains repeated identical three-event chunks, if one hand accounts for more than 60% of events, or if the same hand uses one finger for more than three consecutive occurrences. Consecutive repetitions of the same key by the same hand preserve the same finger assignment, avoiding unrealistic repeated-key finger substitutions. Hand-transition entropy is logged but never used for rejection.

B.5 Family Matching Tolerances

Accepted candidates are deduplicated on their exact event lists, then connected in a compatibility graph in which an edge joins two candidates satisfying every pairwise tolerance in Table B.1; a greedy clique heuristic extracts a mutually compatible family of the requested size. If no family can be matched within tolerance, the generator reports an error for that difficulty bin rather than relaxing the constraint silently, and the experimenter may widen the note range, relax the tolerances, or request a smaller pool.

Table B.1: Pairwise matching tolerances for sequences within the same level pool.

Statistic	Tolerance
H_{norm}	± 0.05
$\overline{d^m}/S$	± 0.04
A_h	± 0.08
B_h	± 0.10
O_{LR}	± 0.05

Saved stimuli are written as a metadata file and a fingering file per sequence, in the same layout used for recorded material, so the playback and cue-delivery tools load generated sequences without special-case handling. The stored target note is accompanied by the resolved physical key identifier from the active profile’s MIDI mapping, so cue delivery addresses the correct calibrated key without relying on a hard-coded key index. A read-only metrics viewer recomputes every component of C_m , C_s , and C_c for saved sequences using the same functions as the generator, so metrics shown during generation and metrics reported later are directly comparable.

B.6 Synchronisation Anchor Self-Checks

The per-trial video–MIDI anchor of Section 2.3.1 is accepted only when two independent self-checks pass. First, the pulse length recovered from the video must match the commanded flash duration, computed from the saved LED-on and LED-off timestamps, within a fixed tolerance; a bright region of the wrong duration is not the mark. Second, the trial offset implied by the rising edge and the offset implied by the falling edge must agree, which fails if either edge was mis-localised by motion or exposure change.

Detection proceeds on the mean brightness of one calibrated key’s pixel region over the opening seconds of the video. A sliding box matched filter of exactly the commanded width selects the window that is bright inside and dark immediately before and after; the search is restricted to ± 0.5 s around the software-expected flash time so that later cue LEDs on the same key cannot be mistaken for the mark, and both edges are then refined on the frame-to-frame brightness derivative. Trials failing either check fall back to the recording start-timestamp estimate and are flagged for frame-level inspection in the review tool, which also serves as a spot audit of the automatic anchors. Each trial stores the anchor it actually used, so any analysis can be traced to its alignment evidence.

B.7 Finger-Matching Decision Rule

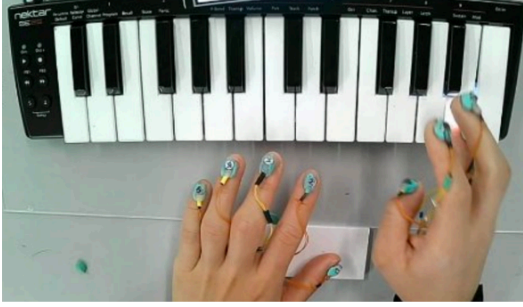
Section 2.3.3 gives the reason for a graded rather than a nearest-fingertip decision; this section records the parameter values it produces. The temperature $\tau = 25$ px is approximately half a white-key width at the working resolution of 1280×720 , so ambiguity is calibrated in units of the response surface: two fingertips within half a key of each other share the mass almost evenly, while a fingertip a full key away is outweighed by a factor of about $e^2 \approx 7.4$. The threshold $\theta = 0.40$ is set below 0.5 so that a bilateral near-tie between the target and one neighbour (approximately 0.5/0.5) is scored correct, reflecting that the camera evidence cannot separate the two; and above $1/3$ so that an event fails whenever any other finger clearly dominates, and also in the rare three-way tie, where the visual evidence carries essentially no information about the acting finger.

Distances are measured to the axis-aligned bounding box of the pressed key’s pixel mask, which keeps the computation stable, with a separate per-event flag recording whether the fingertip also lies inside the exact mask. When several note-on events share a timestamp, candidate pairs are assigned greedily in order of decreasing probability so that one fingertip cannot be credited with two simultaneous notes, while each note’s distribution is still computed over all visible fingertips. This batch capability supports chordal material elsewhere in the platform, so the single-key protocol is an experimental restriction rather than a limit of the matcher. Events with no visible hand in the aligned frame remain unresolved rather than being forced to a label. The complete per-event distribution is stored in the results file, so the correctness criterion can be recomputed exactly under any alternative θ without reprocessing video; the live monitoring view and the offline scoring pipeline call one shared implementation of both the distribution and the threshold rule, so the two paths cannot drift apart.

Figure B.2 shows four of the events whose automatic label the reviewer overrode, the class behind the 4.50% correction rate of Section 3.2.1. Each is a near-tie the softmax resolved towards an adjacent fingertip, which is why the pre-screen was treated as evidence to be checked rather than as a label.

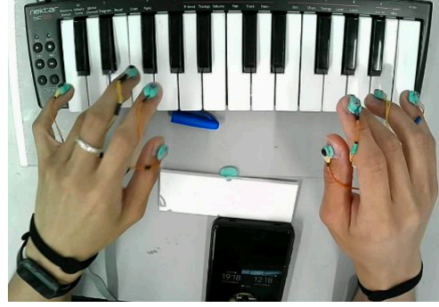
Four keypresses the automatic finger labelling got wrong and the reviewer corrected from the video. In each, the cued finger is hidden behind a neighbour, so both fingertips project onto the same key, the distance-based SoftMax splits between them, and the cued finger falls below the $\theta = 0.40$ threshold.

(a) cued R5 · detector read R2 · $p(R5) = 0.19$



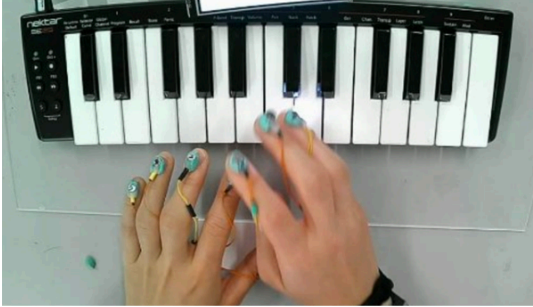
little finger curled in behind the hand

(b) cued L4 · detector read L3 · $p(L4) = 0.28$



ring finger behind the middle finger

(c) cued R5 · detector read R4 · $p(R5) = 0.31$



little finger behind the ring finger

(d) cued R3 · detector read R2 · $p(R3) = 0.38$



middle finger behind the index

Figure B.2: Four keypresses whose automatic finger label was corrected during review. Each panel header gives the cued finger, the finger the detector reported, and the probability the detector assigned to the cued finger.

B.8 Tele-training Measurement Protocol

The transport benchmark issued 1,000 cue-sized probes from the teacher client at a fixed interval after a warm-up, each acknowledged by the learner client, with round-trip time taken from the teacher’s own monotonic clock and application-level loss counted directly. Learner-side software alignment was measured separately in eleven real two-machine sessions: dispatch latency used all 244 unique timed cues, while visual–haptic skew used the 160 cues containing both timestamps. Both measures use the learner’s monotonic clock because the teacher and server clocks were not synchronised to it. These are software timestamps — the instant a repaint is scheduled and the instant a drive command is written — and not physical onsets.

The note-on-to-cue figure quoted in Chapter 3 is a derivation, not a single measurement. Recognition contributes half a tracking period on average, since finger identity is read from the newest completed landmark pass rather than a fresh one (≈ 77 ms for a two-hand 1280×720 pass, measured with both clients sharing one host); one-way transport contributes $RTT/2$; and learner dispatch contributes its measured median. Summing the typical values gives about 55 ms, and the corresponding p95 values give about 170 ms.

Reaction time in the deployment is clocked from the learner’s own cue-ready instant on a single monotonic clock, so transport delay shifts when a cue is presented but never enters the reaction-time interval. The two demonstration sessions were scored by the same automatic pipeline as the main study but without the frame-by-

frame review pass, and used one non-naive operator, unmatched sequence lengths, and no counterbalancing.

Appendix C

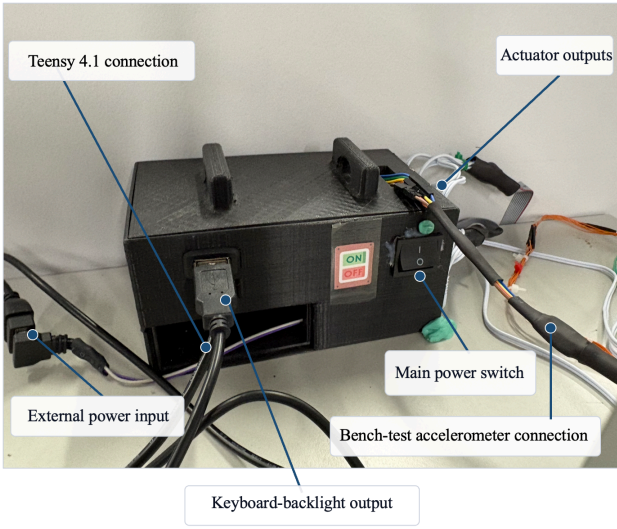
Reproducibility

The values below are read from the working system rather than from design notes, so a discrepancy between this table and the hardware is a fault in one of them and can be checked against the source. The host application and the embedded firmware are at https://github.com/shengjlang/individual_project_2026.

Table C.1: Hardware and software manifest. Motor *ports* are logical channels on the driver boards and correspond one-to-one with the Teensy PWM pins of the same number.

Item	Specification
<i>Control</i>	
Microcontroller	Teensy 4.1
Firmware	<code>haptic-piano v2.10.0</code>
Host link	USB serial; host application in Python (PySide6)
<i>Haptic chain</i>	
Driver boards	$2 \times$ NhaptX v1.0, custom PCBs built around STSPIN240 dual brushed-DC drivers
Finger actuators	Ports 0–9, one per digit: L5→0, L4→1, L3→2, L2→3, L1→4, R1→5, R2→6, R3→7, R4→8, R5→9
Test channels	Port 10 ERM, port 11 LRA (wiring convention, overridable per session)
Drive settings	LRA 224 Hz at 25.1% PWM duty; ERM 1 kHz at 19.6% PWM duty; boot PWM frequency 224 Hz on every channel
<i>Visual cue</i>	
LED strips	$2 \times$ WS2812, GRB, 60 pixels each, data on pins 24 and 29 , driven by WS2812Serial (hardware UART + DMA)
<i>Measurement</i>	
Accelerometers	Up to $3 \times$ LIS3DH on one bit-banged SPI bus (mode 3, ≈ 2.5 MHz): SCK 33 , MOSI 34 , MISO 35 , CS 36/37/38
Keyboard	Nektar SE25, 25 keys, class-compliant USB MIDI
Camera	Unbranded USB webcam, driver-free (UVC), 1080p sensor; mounted overhead and operated at 1280×720 , 30 fps
<i>Power</i>	
3.3 V	Driver-board logic and LIS3DH, from the Teensy 3.3 V pin
5 V	WS2812 strips, from the Teensy Vin pin
9 V	Motor drive rail chopped by PWM, kept off the logic supply. Either a battery or a USB 5 V boost converter set to 9 V (1 A); the latter was used here. For duty $D = \text{amp}/255$, the time-averaged PWM voltage is $9D$ V (2.26 V at the selected LRA setting); the LRA’s vibration is governed mainly by the AC fundamental rather than this mean
Ground	All grounds tied common

(a) External interfaces



(b) Internal electronics

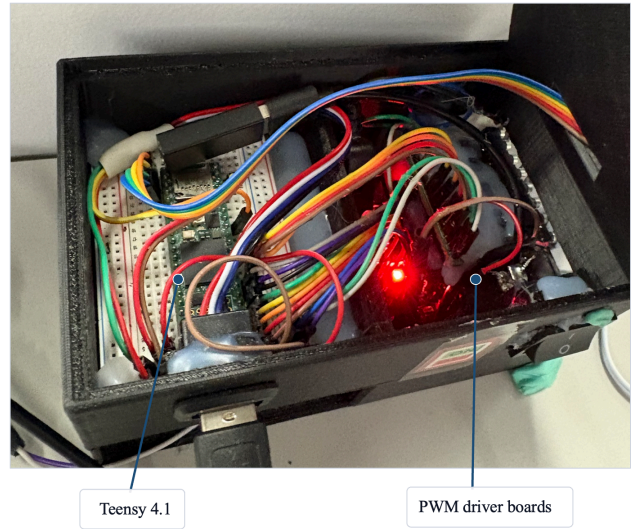


Figure C.1: Integrated 3D-printed control enclosure. (a) External interfaces: Teensy 4.1 connection, keyboard-backlight output, external power input and main switch, actuator outputs, and bench-test accelerometer connection. (b) Internal Teensy 4.1 and PWM driver boards. The enclosure protected the electronics while exposing the interfaces required during bench validation and user sessions; the accelerometer connection was used only for bench testing.

Two constraints are not obvious from the table. Teensy 4.1 inputs are not 5 V tolerant, so every signal line including the LIS3DH bus runs at 3.3 V. The WS2812 data pins are not free choices either: WS2812Serial requires a hardware UART transmit pin, and of the eligible pins the motor PWM outputs and the accelerometer bus already claim the lower ones, which is why the strips sit on 24 and 29.

C.1 Computational Model and Held-out Prediction

Each resolved response was modelled as a softmax choice among the ten candidate fingers. For participant s , condition c , target key k_t , cued finger (h_t^*, d_t^*) , and candidate $f = (h_f, d_f)$,

$$U_{ts}(f) = \rho(h_f, k_t) + w_c[\pi(f) + \pi_s(f)] + \mu_1 I(f = f_{t-1}) + \mu_2 I(h_f = h_{t-1}) \\ + \kappa_{H,c} I(h_f = h_t^*) + \kappa_{D,c} I(d_f = d_t^*) - \gamma_c |d_f - d_t^*|, \\ P(F_t = f) = \frac{\exp U_{ts}(f)}{\sum_{g=1}^{10} \exp U_{ts}(g)}.$$

Condition A has no cue term. M0 removed cue evidence, M1 used a unitary cue, M2 separated hand and digit, M3 added the digit-distance gradient, and M4 freed evidence by cued digit. M3 was the primary interpretable model; M4 tested added flexibility. Penalised maximum-likelihood fitting used the target key, previous completed response, and training-participant data only. Population LOPO held out one participant per fold; Condition-A personalisation was evaluated separately.

Table C.2: Computational-model checks. Log loss is lower when prediction is better; all prediction rows are held out.

Check	Result	Reading
Primary fit	16,195 events; 250 parameters; converged	Descriptive fit only; not evidence of generalisation.
Cue contrasts	Hand: B 5.165, C 8.820; digit sharpness: B 3.584, C 3.744	The C hand estimate is a lower bound: only two C actions used the wrong hand. No ratio is interpreted as an effect size.
LOPO choice prediction	M3 log loss 0.3053; M4 0.2963; multinomial 0.2963; M3–multinomial paired $p = .156$	M3 improved reduced variants, but not generic predictive accuracy.
Homologous-error ranking	M3 captured 35/53, 47/53, and 51/53 in the top 10%, 20%, and 30%; at 20%, M4 captured 51/53 and multinomial 50/53	Risk generalised to unseen participants, but the enrichment was not unique to M3.
Within-hand ranking	38/359 in M3's top 20%	No predictive lift; this broad detected-mismatch class is not the genuine-substitution class used in the confusion analysis.
Event-level RT	Population M3 R^2 : A 0.009, B 0.026, C 0.054	Event prediction was weak; the measured 247 ms B–C difference was not derived by the model.
Condition-A personalisation	Δ log loss = +0.0024, 95% CI $[-0.0039, 0.0087]$, $p = .432$	No improvement in cross-participant prediction.

Figure C.2 links the model structure to the cue-evidence estimates and held-out ranking summarised in Table C.2.

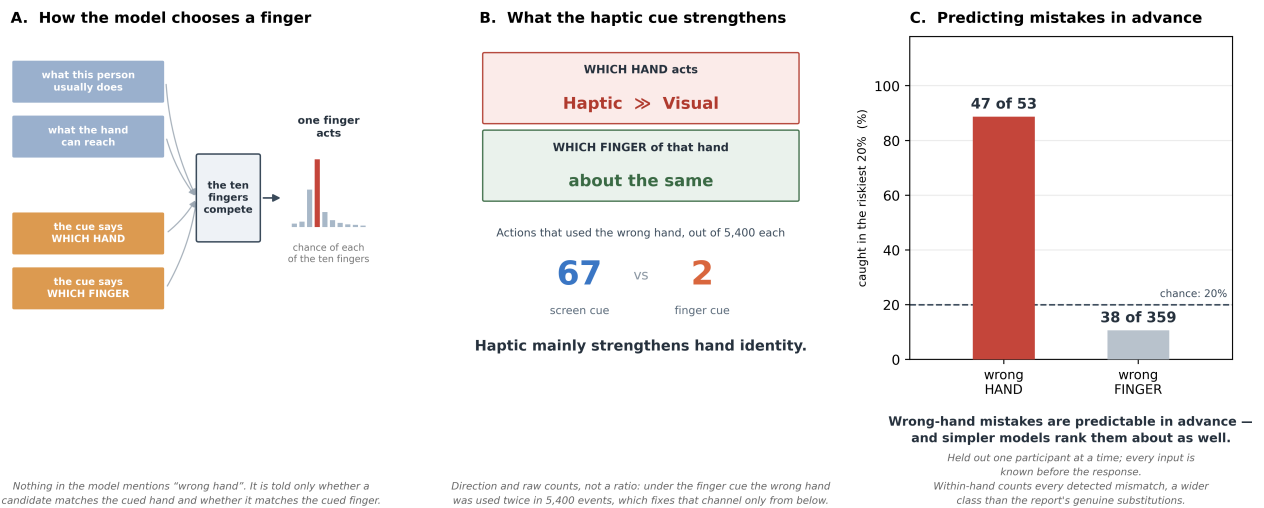


Figure C.2: Computational model and held-out prediction. (A) Ten-finger competition. (B) Estimated cue evidence for hand identity and within-hand digit. (C) Population leave-one-participant-out ranking. The grey bar includes detected mismatches rather than genuine substitutions; generic multinomial prediction ranked wrong-hand risk similarly.

Appendix D

Secondary Behavioural and Deployment Results

This appendix retains five supporting visualisations that provide additional context for Chapter 3 without altering its primary inferences, followed by the unmatched single-operator sessions used only to verify the complete remote path.

D.1 Additional Behavioural Visualisations

Figure D.1 shows the participant trajectories behind the condition-level accuracy summaries.

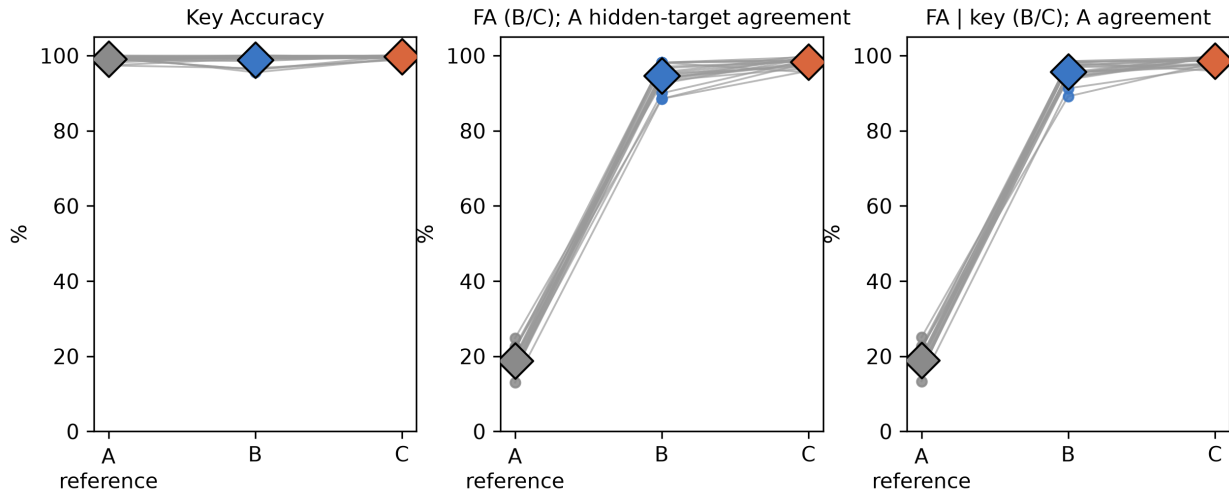


Figure D.1: Accuracy outcomes by condition. Thin lines are participants; diamonds and bars are group means and 95% participant-level bootstrap CIs ($n = 20$). In A, finger measures are agreement with the hidden generated fingering, not accuracy to a supplied cue.

Figure D.2 shows the per-digit Main Finger Accuracy cells summarised in Chapter 3. The outcome is ceiling-bounded and was not entered into the ANOVA; the binomial mixed model fitted to these same cells is Figure 3.8 in the main text.

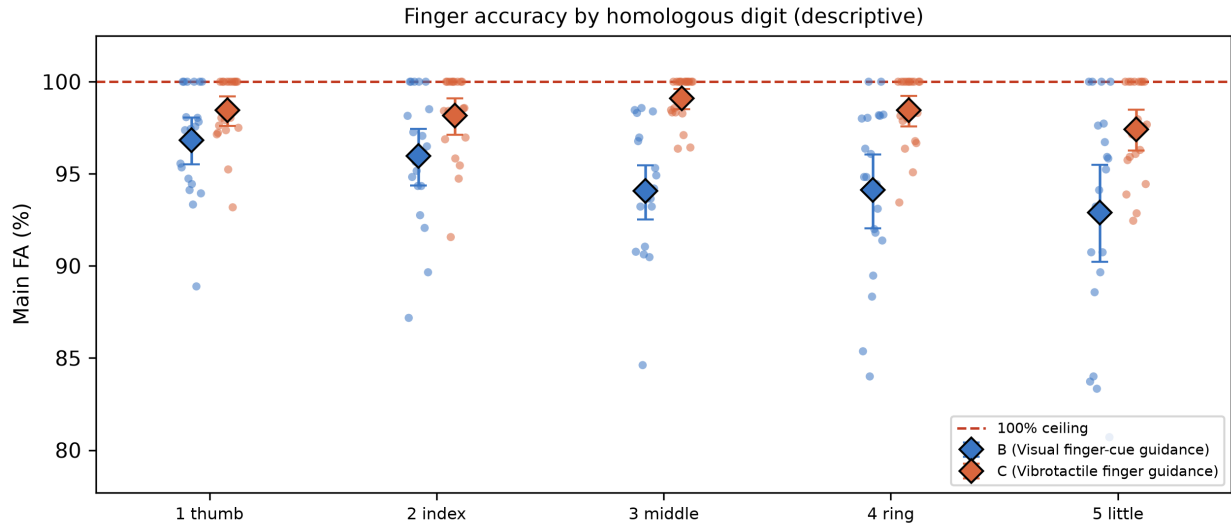


Figure D.2: Main Finger Accuracy by homologous finger. Points are participant cell means; diamonds and bars are group means and 95% participant-level bootstrap CIs ($n = 20$). This ceiling-bounded outcome is descriptive and was not entered into the ANOVA; the model fitted to the same cells instead is Figure 3.8.

Figure D.3 shows participant-level rates for correct-key events assigned to a different finger. In B and C this is a cueing error; in A it is disagreement with the hidden generated reference.

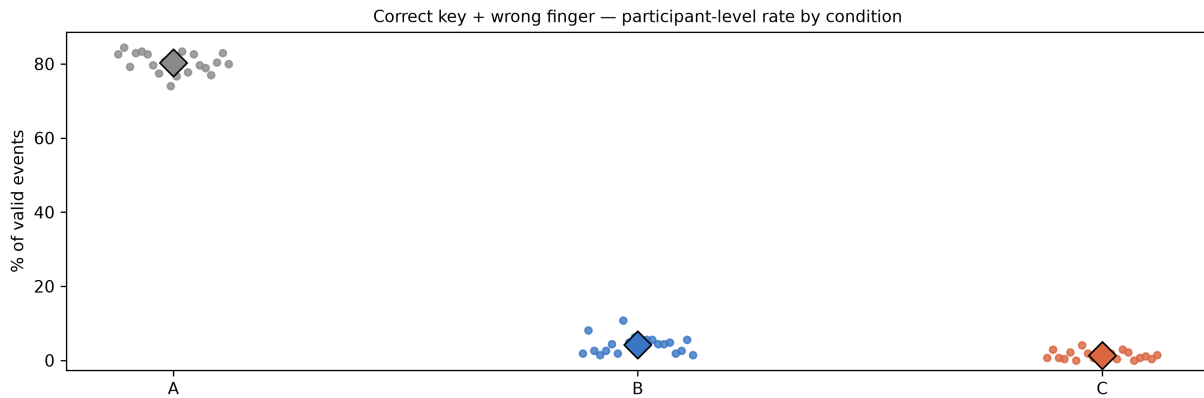


Figure D.3: Participant-level proportion of valid events with the correct key but a different finger. In B and C, “different” means different from the supplied cue; in A, it means different from the hidden generated reference and is not an error against an instructed finger. Points are participants; diamonds and bars are group means and 95% participant-level bootstrap CIs ($n = 20$).

Figure D.4 gives the observed difficulty-aligned trajectories that complement the controlled Condition $\times$ Repetition analysis in Chapter 3.

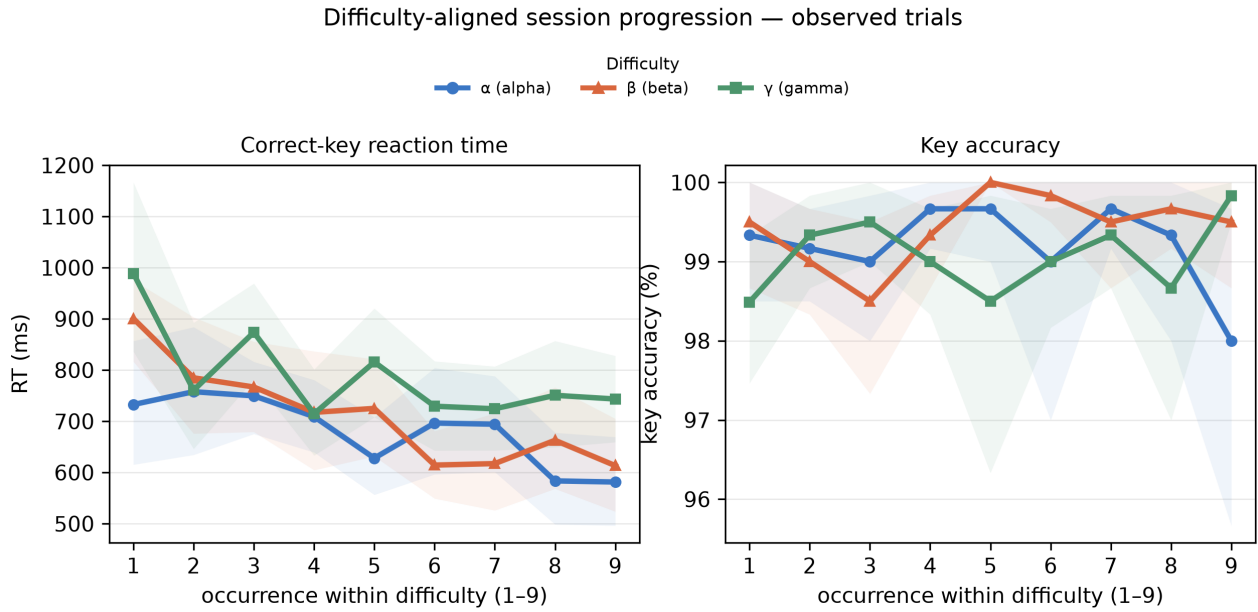


Figure D.4: Difficulty-aligned within-session progression across all A/B/C trials. Within each participant and difficulty, trials are sorted by actual session order and relabelled occurrence 1–9. Marked lines are group means and shaded bands are 95% participant-level bootstrap CIs ($n = 20$). Panels show observed correct-key reaction time and key accuracy. Because condition composition varies across occurrences, the trajectories are descriptive; condition-adjusted values are exported but not plotted.

Figure D.5 shows the participant-level hand preference underlying the Condition A summary in Chapter 3.

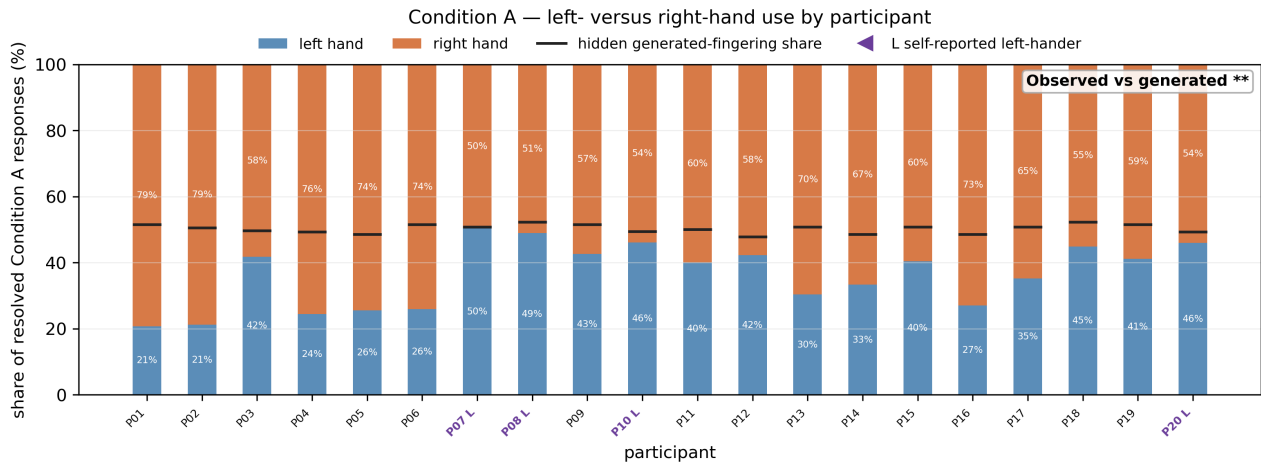


Figure D.5: Left- versus right-hand share of resolved Condition A presses by participant. Horizontal rules mark the hidden generated-fingering share; purple “L” labels mark the four self-reported left-handers. The overall observed-versus-generated annotation uses a two-sided paired t -test ($*p < 0.05$, $**p < 0.01$); comparisons by handedness remain descriptive ($n = 20$).

D.2 Tele-training Demonstration Sessions

Two single-operator sessions exercised the complete remote path. Remote visual guidance yielded 13 correct actions in 16 events at a mean of 951 ms; remote vibrotactile guidance yielded 44 of 47 at 500 ms. These unmatched

sessions used one non-naive operator, no counterbalancing, and automatic scoring without frame-by-frame review. They verify end-to-end delivery but do not estimate a remote behavioural effect.

Appendix E

AI Use Statement

Artificial intelligence tools were used in a supportive capacity during this project. During report preparation, they assisted with language polishing, grammar correction, sentence restructuring, refinement of academic phrasing, and section organisation. They were also used to help clarify and shorten technical descriptions, figure captions, and explanations of results. Selection and verification of the literature, interpretation of sources and findings, and decisions about the direction of the work remained the author's responsibility.

AI tools additionally provided suggestions during debugging, helped clarify technical problems and code comments, and supported improvements to code readability and organisation. The author reviewed and verified suggestions before use. The central intellectual and practical work—including problem formulation, study and model design, implementation, parameter selection, experimental evaluation, interpretation of results, and critical discussion—was carried out by the author.